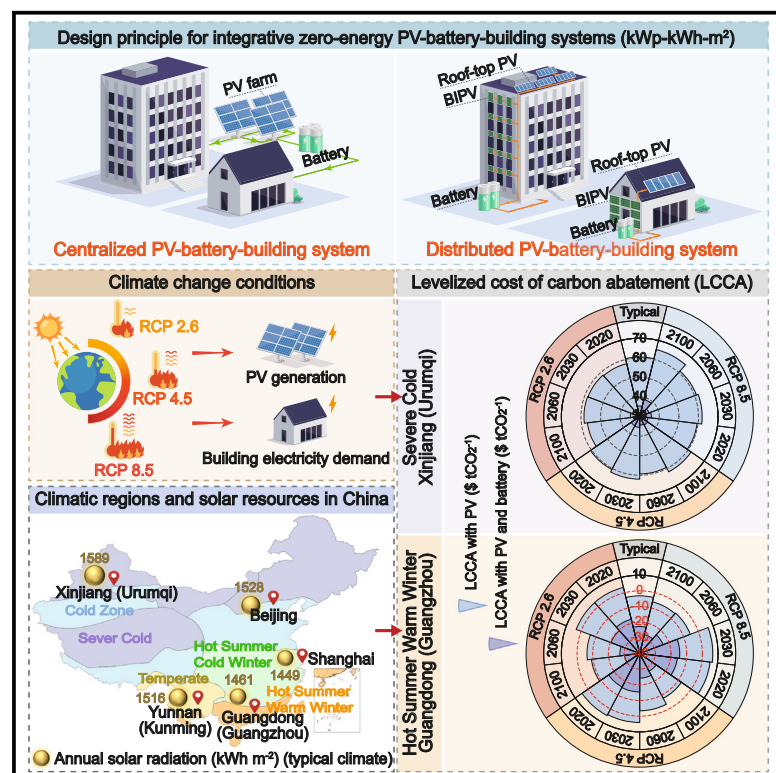


Energy-resilient climate adaptation using a tailored life-cycle integrative design approach for national carbon abatement

Graphical abstract



Authors

Lu Zhou, Yuekuan Zhou

Correspondence

yuekuan.zhou@outlook.com

In brief

This study proposes a tailored “kWp-kWh-m²” design approach to achieve renewable self-sufficiency and avoid battery oversizing in both centralized and distributed zero-energy building systems. Zhou et al. quantify and compare life cycle and economic-environmental performance across diverse climate conditions and geographic areas in China for guiding sustainability transformations.

Highlights

- A tailored “kWp-kWh-m²” design approach for zero-energy building systems
- Optimal battery sizing for centralized and distributed prosumer-storage scenarios
- Economic investment and decision making for zero-carbon transformations
- Zero-energy transitions with guidelines and strategic investment are suggested



Article

Energy-resilient climate adaptation using a tailored life-cycle integrative design approach for national carbon abatement

Lu Zhou¹ and Yuekuan Zhou^{1,2,3,4,5,*}

¹Sustainable Energy and Environment Thrust, Function Hub, The Hong Kong University of Science and Technology (Guangzhou), Nansha, Guangzhou, Guangdong 511400, China

²Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong SAR, China

³HKUST Shenzhen-Hong Kong Collaborative Innovation Research Institute, Futian, Shenzhen 518048, China

⁴Division of Emerging Interdisciplinary Areas, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong SAR, China

⁵Lead contact

*Correspondence: yuekuan.zhou@outlook.com

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SUMMARY

Greenhouse gas escalation and global warming impose substantial challenges for adapting to and mitigating climate change while ensuring energy resilience. However, uncertainties in supply-demand dynamics and climate-adaptive resilience remain unclear for electrified and integrative photovoltaic (PV)-battery-building systems. Here, following zero-energy building design principles and a U-value/M-value battery sizing method based on maximum marginal benefits, a tailored “kWp-kWh-m²” design approach is applied with intrinsic relationships of prosumer-storage to achieve renewable self-sufficiency and avoid battery oversizing in both centralized and distributed forms. Economic-environmental assessments for zero-energy transformations with the integrative PV-battery systems are conducted across diverse climate-change conditions and geographic areas in China, in terms of levelized costs of storage, net present values, decarbonization potentials, and policy incentives. This research provides guidelines for zero-energy transitions from optimal system design, provincial-level system configurations, and performance evaluation, aiming to guide strategic investment and targeted policy interventions.

INTRODUCTION

The Paris Agreement underscores the global efforts to combat climate change through the enforcement of nationally determined contributions (NDCs) and long-term strategies (LTSs).¹ In response, China has committed to peaking its carbon dioxide emissions before 2030 and achieving carbon neutrality by 2060.² The building sector plays a significant role in this context, contributing approximately 37% of total carbon emissions, with around 75% of these emissions arising from the operational phase, primarily driven by electricity consumption.^{3,4} Hence, mitigating carbon emissions in the building sector during the operational stage emerges as a pivotal pathway toward sustainable transitions.

As highlighted in NDCs, the extensive deployment of renewables is recognized as a pivotal solution for driving sustainability transitions.⁵ The low-carbon energy transition is essential for combating climate change by improving resource utilization, reducing environmental emissions, and generating employment.⁶ Within the decarbonization efforts of the building sector, renewable technologies such as roof-top photovoltaic (PV) systems have proved to be vital in effectively reducing building-

related carbon emissions.⁷ One innovative concept that has gained prominence in this context is the zero-energy building (ZEB), which refers to a building energy system that generates as much renewable energy as it consumes over a given period. This has gained prominence through policy implementations and research efforts and is regarded as a significant contributor to the decarbonization of the building sector.⁸ However, challenges from scaling renewable generation and addressing supply-demand mismatches within the ZEB framework necessitate the integration of energy-storage solutions. The energy-storage techniques are considered to be an effective solution for achieving deep decarbonization while enhancing system performance in terms of energy flexibility and resilience.^{9,10} However, due to the conflicts between large battery capacity for renewable storage and the increased investment costs, optimal battery sizing within the ZEB system is crucial to achieve the trade-off with the balance between renewable storage and initial investment cost. Additionally, dynamic climate conditions significantly influence optimal battery sizing due to the variations in renewable generation and load patterns.¹¹ To identify the optimal renewable-battery framework, numerous studies have focused on optimizing battery sizing for renewable-integrated systems.



Methodologies have been extensively applied to optimize battery capacity within renewable-integrated energy systems, such as genetic algorithm (GA), dynamic programming (DP), general algebraic modeling system (GAMS), and linear programming (LP) methods.¹² For application scenarios, Peters et al.¹³ focused on the battery capacity configurations for constant-load scenarios. Mulleriyawage et al.¹⁴ optimized battery capacity in small-scale PV-integrated residential buildings, and Elkazaz et al.¹⁵ proposed a battery-sizing methodology for a community battery energy storage system (CBESS). However, most existing literature primarily focused on either optimal technical performance¹⁶ or economic performance^{17,18} without identifying and determining the trade-offs between technical-economic feasibility. There is a lack of literature that provides comprehensive battery-sizing methods for diverse application scenarios, particularly combining techno-economic-environmental impacts under varying climate-change conditions. Moreover, under climate-change conditions, disparities and fluctuations in power supply,^{19,20} electricity demand,^{21,22} and system design principles^{23,24} across diverse building energy systems significantly affect the system performance from both economic and environmental aspects.²⁵ Current literature failed to provide a comprehensive analysis of climate-change resilience across various renewable application scenarios, which is critical for decision making by both policy makers and technology investors.

Globally, significant efforts such as the promotion of policy combinations have been made to drive sustainability transitions and achieve notable carbon reductions.²⁶ In China, long-term sustainability faces challenges from socioeconomic and environmental uncertainties.²⁷ Aligned with the national sustainable transition goals of achieving carbon peak and carbon neutrality, provinces in China have implemented a combination of policies, investment, and financial incentives.^{28–30} Nevertheless, the diverse climate regions with distinct renewable resources and weather conditions impose significant challenges in charting the regional transition roadmaps toward sustainability transformations. Key questions remain unanswered; e.g., what are the exact capacities for renewables that should be installed? How about the installed battery storage capacities? How about the economic-ecological-policy feasibility? Most critically, both the vulnerability and resilience of the building energy system under varying climate-change conditions remain unclear, imposing great challenges for effective government policy guidance. Understanding these factors is crucial for shaping sustainable energy transition pathways for China under uncertain climate-change conditions.

Based on the literature review, several scientific gaps can be identified.

- (1) The growing instability of renewable sources and stochasticity of building electricity loads from climate change or extreme human activities impose significant challenges for demand response and on-site renewable consumptions, leading to challenges for energy-resilient system establishment and zero-energy transformations. Integrative energy-system design can effectively combat climate change and survive power systems with high energy resilience. However, the system design principles of integra-

tive renewable-battery-demand building systems toward zero-energy transitions under climate change are still unclear, considering diverse renewable supply paradigms and building energy consumption patterns.

- (2) Economic-environmental performances of diverse renewable energy systems highly determine the decision making of various renewable system investors, while the impact of future climate change on both energy supplies and energy demands is still unclear. Furthermore, the levelized value over the lifetime of various renewable systems has not been studied and evaluated yet, leading to uncertainties in diverse renewable systems deployment for stakeholders, such as investors, operators, and even consumers.
- (3) Considering various meteorological parameters for five different climates and carbon neutrality goals in 2060 for the Chinese government, renewable-battery capacity design in both centralized and distributed energy systems is essential, while challenges are from uneven regional distribution of renewable resources, distinctive building demand profiles, and significant disparities in building stocks across China. However, the current literature fails to provide guidelines and associated techno-economic-environmental performances across different geographical locations in China.

To address the above research gaps, this research aims to answer questions by the following means:

- (1) Following the intrinsic relationship among the integrative renewable-battery-building system, a tailored approach is applied to guide the capacity design with quantifiable “kWp-kWh-m²” indicators. “U value” from existing literature and self-developed “M-value” methods with the balance between flexibility-economic-environmental performances are applied to guide the battery capacity sizing for both centralized and distributed renewable supply systems.
- (2) To guide investments in optimal renewable-battery-building systems under zero-energy-system configurations with future climate change, a comprehensive economic-environmental analysis is provided with net-present value, decarbonization potentials, and levelized cost of carbon abatement (LCCA), establishing a robust database for large-scale systems analysis.
- (3) This research explores the required capacity of building-integrated PV (BIPV)/PV and battery systems for achieving ZEB systems across five regions with distinct climate conditions in China, with provincial analysis for economic-environmental implications. Meanwhile, the role of batteries in facilitating the transition toward terawatt levels of PV deployment is emphasized, in terms of economic and ecological aspects.

The research aims to quantify the impact of climatic change conditions on system design principles, configurations, and performance evaluation. This study is organized as follows. The “Results” section presents the quantitative findings and discussions, followed by a summary in the “Discussion” section.

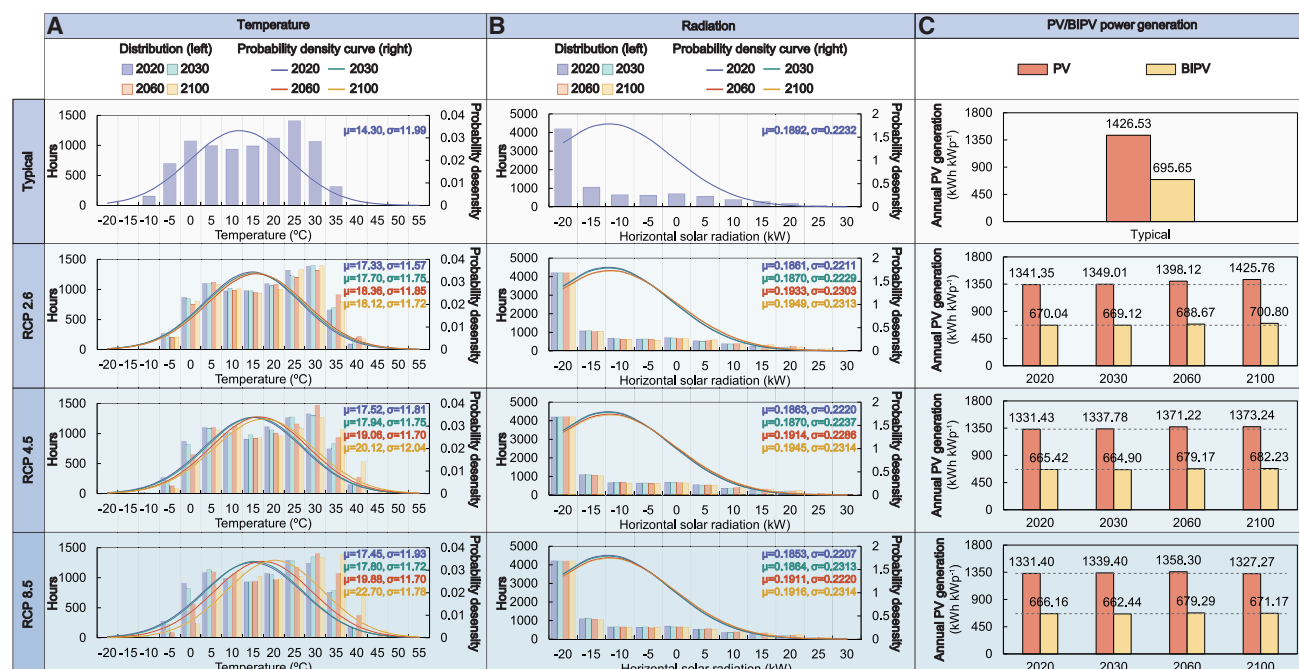


Figure 1. Future climate change

Future climate change under historical typical year and RCP 2.6/RCP 4.5/RCP 8.5 for 2020/2030/2060/2100: (A) temperature; (B) horizontal solar radiation; (C) PV/BIPV power generation. The installation angles of the roof-top PV/PV farm follow the local attitude to maximize the solar exposure, and the BIPV is installed evenly on all four facades of the building. The electric performance of the installed solar PV module is listed in Table S1.

The “Experimental procedures” section outlines the methodologies regarding the simulation platform and battery sizing and provides detailed formulations for applied criteria. The results provide adaptive guidelines for ZEB design and demonstrate the potential benefits as well as the challenges associated with clean-energy supply transitions under climate-change conditions.

RESULTS

Impact of climate change on PV energy supply

Aligned with global efforts to transition from fossil fuels to sustainable energy supply, PV technology has garnered increasing attention. PV systems offer the ability to provide clean, cost-effective electricity and exhibit versatility across a wide range of applications, from distributed roof-top/building-integrated systems to large-scale centralized solar farms. However, PV generation is significantly affected by climate change due to its inherent dependence on various climatic factors, such as solar radiation, temperature, cloud cover, precipitation, and extreme weather events. Consequently, dynamic PV system performances under climate-change conditions are essential to guide strategic investments. In this section, future climate-change models have been developed to predict meteorological parameters (with the assumption of temperature rise in 2100 lower than 2°C [representative concentration pathway (RCP) 2.6], 2°C–3°C [RCP 4.5], and 3°C–5°C [RCP 8.5]), using single-variable model (maximum monthly temperature model³¹). Afterward,

the impact of climate change on PV generation is investigated, using Beijing as an example.

To examine the impact of climate change on PV generation, Figure 1 demonstrates the annual distribution and probability curves of dry bulb temperature and horizontal radiation under various climate-change scenarios. The data reveals that the ambient temperature steadily increases as the climate-change scenario transits from the historical typical year to the more severe RCP 2.6, RCP 4.5, and RCP 8.5 pathways. As shown in Figure 1A, the mean temperature rises from 14.30°C in the historical typical year to 17.33°C under RCP 2.6 in 2020, to 17.52°C under RCP 4.5 in 2020, and then to 17.45°C under RCP 8.5 in 2020. Furthermore, under all RCPs, the temperature significantly rises over time from 2020 to 2100, and the increase is more obvious in RCP 8.5, with the mean temperature climbing from 17.45°C in 2020 to 22.70°C in 2100 by 5.25°C. These temperature increases are driven by higher concentrations of greenhouse gases in the atmosphere, underscoring the profound impacts of climate change on global warming and the urgency of developing effective solutions for climate-change mitigation.

Regarding horizontal solar radiation, it is notable that the higher RCPs lead to lower solar radiation. This can be observed in the decrease of mean solar horizontal radiation values from the historical typical year to RCP 2.6, RCP 4.5, and RCP 8.5. Take the year 2020 as an example, the mean horizontal radiation decreases from 0.1892 kW in the historical typical year to 0.1861 kW in RCP 2.6, 0.1863 kW in RCP 4.5, and 0.1853 kW in RCP 8.5. This reduction may be primarily due to the increased aerosol emissions under higher RCPs, which diminish the

amount of solar radiation that reaches the Earth's surface. However, under the same RCP, the solar radiation tends to increase over time, together with an increase in standard deviation. Specifically, the mean solar horizontal radiation under RCP 2.6 rises from 0.1861 kW in 2020 to 0.1949 kW in 2100, together with the increase of standard deviation from 0.2211 to 0.2313, indicating greater variability in solar radiation as time progresses. The increase in radiation can be attributed to the changes in atmospheric composition and climate patterns, as well as human interventions in improving the transmission of solar radiation through the atmosphere.

Based on the evolution of ambient temperature and solar radiation under climate change, [Figure 1C](#) demonstrates the variations of annual PV generation (kWh kWp^{-1}) under different climate-change scenarios. It can be noticed that the higher RCPs lead to a lower PV generation. For instance, compared to the historical typical year with annual PV generation at $1,426.53 \text{ kWh kWp}^{-1}$, the value decreases to $1,341.35 \text{ kWh kWp}^{-1}$ for RCP 2.6 in 2020, and further declines to $1,331.43 \text{ kWh kWp}^{-1}$ for RCP 4.5 in 2020 and $1,331.40 \text{ kWh kWp}^{-1}$ for RCP 8.5 in 2020. This reduction in PV generation is primarily attributed to the increased ambient temperatures (as shown in [Figure 1A](#)) under higher RCP scenarios, which leads to a reduction in PV efficiency. Regarding the time-series analysis under climate change, the annual PV energy generation increases over time from 2020 to 2100 under all RCP scenarios, except for RCP 8.5. Under RCP 8.5, the annual PV generation increases from $1,331.40 \text{ kWh kWp}^{-1}$ in 2020, to $1,339.40 \text{ kWh kWp}^{-1}$ in 2030 and $1,358.30 \text{ kWh kWp}^{-1}$ in 2060, but then decreases to $1,327.27 \text{ kWh kWp}^{-1}$ in 2100. The initial increases are due to the rise in solar radiation levels over time, as observed in the analysis in [Figure 1B](#), while the PV generation decrease from 2060 to 2100 under RCP 8.5 is caused by the negative effects of substantial temperature rise for reducing PV efficiency, which overwhelms the increase in solar radiation. Last but not least, the energy generation of roof-top PV/PV farm systems is significantly higher than that of BIPV systems. According to the results, the total annual energy generation of BIPVs is only 48%–51% of that of roof-top PV/PV farms, which can be attributed to the suboptimal orientation and tilted angles of BIPVs due to building design constraints. Nevertheless, given its dual functionality in terms of space utilization and on-site clean-energy supply without power transmission losses and fast demand responses, BIPV still remains a promising technique for sustainable building systems even though the power generation is slightly lower than roof-top PV/PV farms.

Impact of climate change on building energy demand

Considering the large proportion of carbon emission from buildings at 37%³ and that 75% originates from the operational stage due to energy consumptions,^{3,4} this section aims to provide an evolution of building electricity demands and evaluate carbon reduction potentials under climate change. Through a rigorous approach combining system modeling, load simulation, and equipment resizing, this section investigates the impact of climate change on the dynamic electricity demands of two distinct building types (i.e., residential and office buildings) and selects Beijing as an example. The results can quantify the po-

tential impact of climate change on electricity consumption, thereby contributing to the broader decarbonization within the built environment transition toward sustainability.

[Figure 2](#) demonstrates electricity demands and their compositions for residential buildings and office buildings located in Beijing under climate-change conditions. Generally, under all RCPs, the building electricity demand steadily decreases as time moves from 2020 to 2100, and the decline becomes more significant under the high-RCP scenario (RCP 8.5). For example, for office buildings, the annual electricity demand decreases from $153.56 \text{ kWh m}^{-2}$ to $151.18 \text{ kWh m}^{-2}$, by 1.55% in RCP 2.6, and from $155.42 \text{ kWh m}^{-2}$ to $148.90 \text{ kWh m}^{-2}$, by 4.19% in RCP 8.5. This decrease can be attributed to the rising temperatures from climate change, which reduce the building space heating demand and overwhelm the effects of increased cooling demands in the heating-dominated region (Beijing). Moreover, it is worth noting that the demand decline over time is more obvious in residential buildings than in office buildings, which decreases from $188.13 \text{ kWh m}^{-2}$ in 2020 to $167.79 \text{ kWh m}^{-2}$ in 2100, by 10.81% under RCP 8.5. This is because climate change with rising temperatures can significantly reduce the heating demands but increase the cooling demands, and the residential buildings show a higher ratio of heating demand to total cooling demand (about 3:1) than that of office buildings (only about 1.3:1). Overall, the consistent decrease of electricity demand over time under all RCPs demonstrates the positive effects of climate change on building demand reduction, indicating the potential for combining effective solutions to optimize the building energy system and accelerate the low-carbon transitions under climate-change conditions. However, it is worth noting that the impact of climate change on building electricity demands is highly dependent on the specific geographic location and climatic conditions (i.e., decrease in building electricity demands in the heating-dominated regions but increase in building electricity demands in the cooling-dominated regions), and for the specific demand variations under climate change for five regions with distinct climates in China, please refer to [Figures S1](#) and [S2](#).

Impact of climate change on battery capacity for net-zero-energy system

The fluctuating climatic factors with climate change have a substantial impact on the performance of both PV systems and building energy systems, highlighting the crucial need for incorporating climate-resilient design and adaptation strategies in the development of renewable-building-energy systems under varying climate-change conditions. With the drastic acceleration in PV capacity, the key obstacle to clean-energy transitions is the low energy utilization ratio induced by the mismatch between the intermittent renewable generations and stochastic energy demands. To tackle this issue, the electrochemical battery is considered as a promising energy-storage technique, offering high energy efficiency and fast charging/discharging response. However, the integration of batteries imposes challenges in terms of high initial investment, degradation costs, and embodied carbon emissions. Therefore, accurate battery sizing is crucial for achieving optimal economic-environmental performance and avoiding fortune/material waste. This section

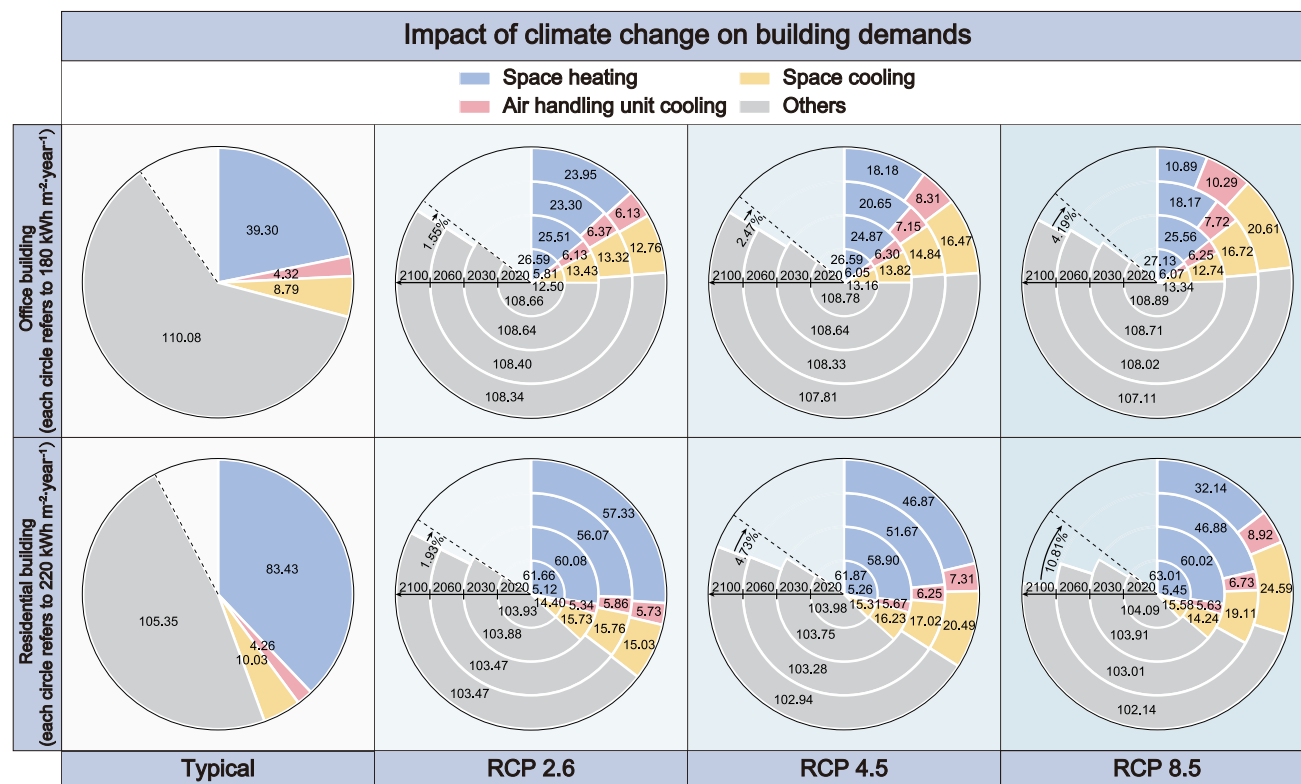


Figure 2. Impact of climate change on building demands

Impact of climate change on annual cooling/heating loads for office and residential buildings under historical typical year, RCP 2.6, RCP 4.5, and RCP 8.5. For the air handling unit cooling (AC) and space cooling (SC) chillers resizing under climate-change conditions, please refer to Tables S2–S6.

explores the optimal battery capacity under both centralized and distributed PV-supplied scenarios to support the PV-building systems with a compromise between the flexibility-cost-environmental impacts.

Considering intermittent PV generation, stochastic building demand, and dynamic device degradation under varying climate-change conditions, Figure 3 demonstrates the optimal battery capacity sizing for both centralized PV power supply and distributed PV power supply systems in both office and residential buildings, respectively. Generally, the optimal capacity ranges from 3.75–4 kWh kWp^{−1} for centralized PV systems, 1.75–2 kWh kWp^{−1} for distributed PV systems in office buildings, and 2.5–2.75 kWh kWp^{−1} for distributed PV systems in residential buildings. The upper part of each chart shows the uniformity factor (U value) (Figure 3B) and matching factor (M value) (Figure 3C) as functions of battery capacity. The lower part of each chart demonstrates the first derivative of these functions. The optimal battery capacity is determined by the point with the highest increase in the uniformity or matching factor, which aims to maximize the marginal efficiency of battery integration, ensuring the highest improvement in load matching while achieving the compromise between flexibility-cost-environmental impacts.

As shown in Figure 3, it can be noted that the centralized PV-supplied scenario requires a larger optimal battery capacity (3.75–4 kWh kWp^{−1}) compared to the distributed PV systems

(1.75–2.75 kWh kWp^{−1}). This can be attributed to the higher requirement for larger battery storages to ensure a stable and consistent power supply across diverse load profiles without the direct renewable self-consumptions in the centralized PV-supplied scenario. Furthermore, according to Figure 3A, the residential building requires a larger battery capacity (2.5–2.75 kWh kWp^{−1}) than the office building (1.75–2 kWh kWp^{−1}). This is due to the lower matching level between BIPV/PV generation and building electricity demand in the residential building (i.e., the peak solar power supply at daytime does not match with the peak energy demands at morning and night) than in the office building (i.e., the peak solar power supply at daytime matches the peak energy demands at daytime).

Impact of climate change on the integrative design of PV-battery-building system for zero-energy transition

ZEB system design represents a sustainable building design approach that aims to balance the building's electricity consumption and renewable energy generation over the investigated period (e.g., 1 year or the whole lifetime). The ZEB framework serves as a benchmark for transitioning toward energy-efficient and environmentally friendly buildings. Based on the above analysis of climate-change impacts on PV generations, building demands, and optimal battery capacity sizing, the integrative design of a PV-battery-building system with a relationship among PV capacity, optimal battery capacity, and demand

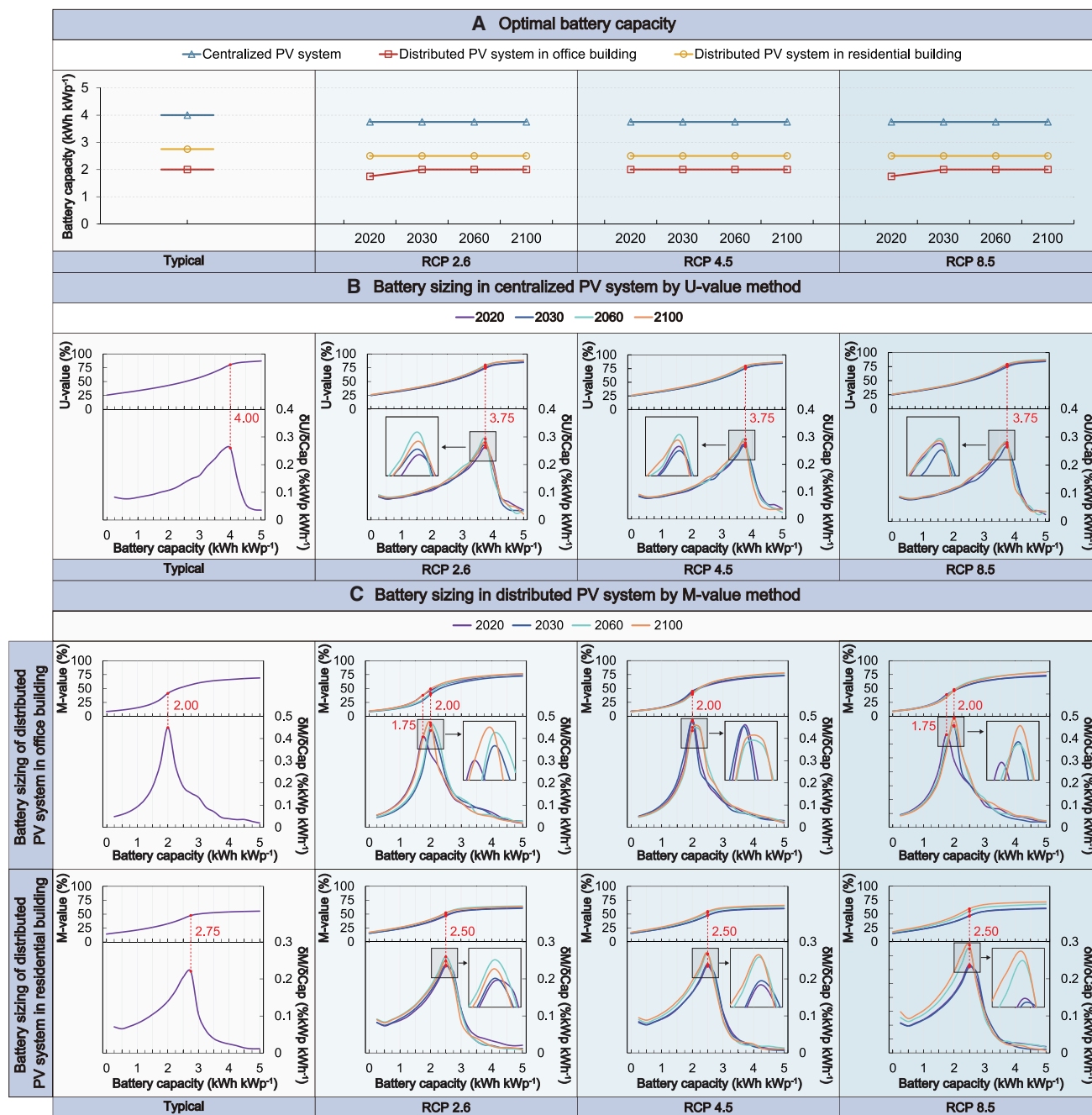


Figure 3. Optimal battery capacity and battery sizing procedures

(A) Optimal battery capacity; (B) battery sizing for centralized PV-battery energy supply scenario with uniformity-value (U value) method; (C) battery sizing for distributed PV-battery energy supply scenario with matching-value (M-value) method. For details of the U-value method and M-value method, refer to section "Experimental procedures." For the battery-degradation model, refer to Note S1, Figures S3–S4, and Table S7. The comparative analysis with traditional battery sizing methodologies is shown in Note S5 to highlight the superiorities of the developed methods.

profile (kWp-kWh-m⁻²) can be established, providing rigorous system design principles to guide the ZEB design on the integrative PV-building battery systems under varying climate-change conditions.

Figure 4 illustrates the impact of climate change on the optimal design of the integrative PV-battery-building

systems. Notably, the required PV and battery capacities per unit area (kWp-m², kWh-m²) decrease over time from 2020 to 2100 across all RCPs. For instance, under RCP 2.6, from 2020 to 2100, the required PV-battery capacity decreases from 0.1238 kWp-0.4644 kWh-m² to 0.1146 kWp-0.4298 kWh-m² (for centralized PVs in the office building),

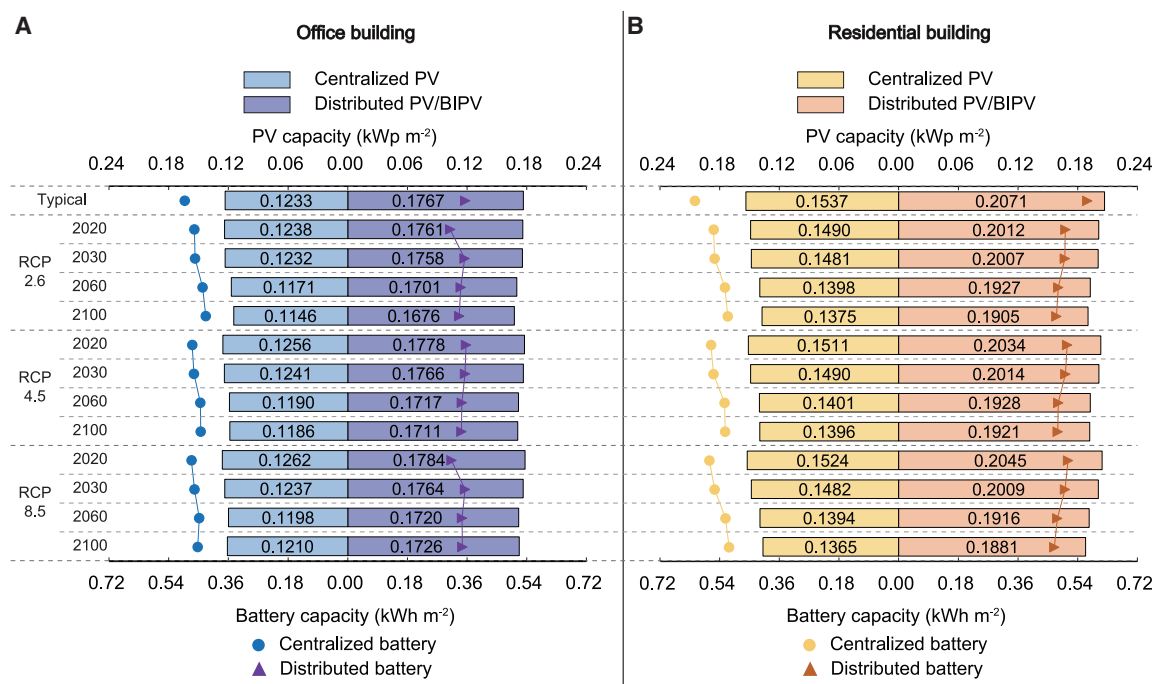


Figure 4. Design principles of the integrative PV-battery-building system for four scenarios

Comparisons of the integrative PV-battery-building system design (kWp-kWh-m²) for both centralized and distributed PV-supply systems: (A) office building; (B) residential building under historical typical year, RCP 2.6, RCP 4.5, and RCP 8.5, respectively.

from 0.1761 kWp-0.3082 kWh-m² to 0.1676 kWp-0.3353 kWh-m² (for distributed PV/BIPV in the office building), from 0.1490 kWp-0.5586 kWh-m² to 0.1375 Wp-0.5156 kWh-m² (for centralized PVs in the residential building), and from 0.2012 kWp-0.5031 kWh-m² to 0.1905 kWp-0.4763 kWh-m² (for distributed PV/BIPV in the residential building). The underlying reason is the climate-change impact on reducing the building electricity demands and increasing the PV generations, as discussed in Figure 1 and Figure 2. This decrease indicates the financial viability and accessibility of ZEB transformations, encouraging the wider adoption of the designed PV-battery system. Moreover, compared to the distributed PV-battery energy system, the centralized PV-battery energy system requires smaller PV capacities but larger battery capacities; for example, in the office building under RCP 2.6 in 2020, the required PV-battery capacity is 0.1238 kWp-0.4644 kWh-m² for centralized PVs and 0.1761 kWp-0.3082 kWh-m² for distributed PV/BIPVs. This can be attributed to the higher PV generation potentials and larger battery storage requirement for centralized PV systems, as highlighted in Figure 1 and Figure 3.

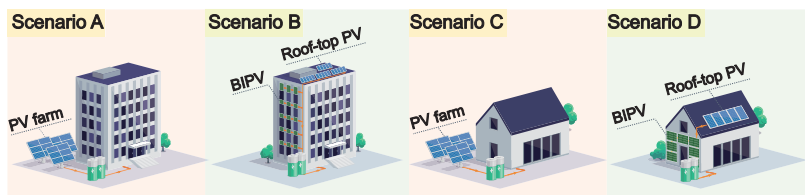
Life-cycle economic-environmental performance

Quantification of economic-environmental performances under climate-change conditions is crucial for exploring the potential of ZEB systems in promoting sustainability transformations. In order to provide rigorous economic and ecological evaluations to guide both benefit-oriented investors and green development-targeted policy makers, this section quantifies the eco-

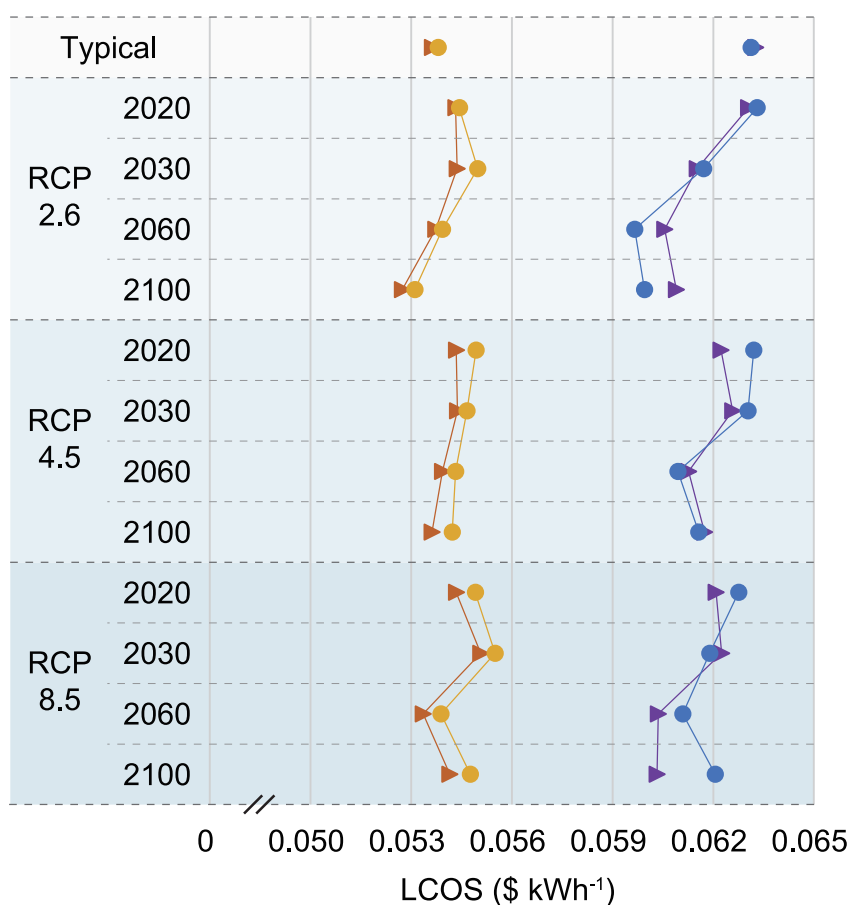
nomic-environmental performances based on the ZEB design in various applications.

In this section, four key performance criteria are analyzed: levelized cost of storage (LCOS), net present value (NPV), decarbonization potential, and LCCA. Specifically, LCOS assesses the economic feasibility and competitiveness of energy-storage technologies compared to other power-supply alternatives. NPV provides a metric for return-on-investment evaluation, enabling informed decision making on energy investments. In terms of environmental aspects, decarbonization potential measures the effectiveness of PV-battery technology in reducing carbon emissions in the energy sector, informing strategic decisions on sustainability transitions. LCCA represents the economic-environmental competitiveness of the integrative PV-battery system within the ZEB context, allowing comparisons with current carbon prices to evaluate the feasibility and practicability of carbon-related policies. This is also crucial in guiding the establishment of a robust carbon-market framework and enabling effective carbon-neutrality strategies in China.

Figure 5 compares the LCOS under four scenarios for both centralized and distributed PV-battery power systems under different climate-change conditions. As shown in Figure 5, considering varying climate-change conditions, it can be noted that the LCOS exhibits a consistent decline over time from 2020 to 2100 across all RCP scenarios. For instance, under RCP 2.6, the LCOS decreases from \$0.633 kWh⁻¹ to \$0.0599 kWh⁻¹ in scenario A, from \$0.0630 kWh⁻¹ to \$0.0609 kWh⁻¹ in scenario B, from \$0.0544 kWh⁻¹ to



- Scenario A: Centralized PV-battery in office building
- ▲ Scenario B: Distributed PV-battery in office building
- Scenario C: Centralized PV-battery in residential building
- ▲ Scenario D: Distributed PV-battery in residential building



\$0.0531 kWh^{-1} in scenario C, and from \$0.0543 kWh^{-1} to \$0.0527 kWh^{-1} in scenario D. This downward trend is primarily attributed to the increasing PV energy generation capacity, which elevates the battery's depth of discharge (DoD) and consequently reduces the LCOS. The declining LCOS signifies the long-term feasibility of investment, thereby incentivizing the widespread integration of renewable energy into energy systems. This also presents an attractive investment opportunity for stakeholders seeking to transition toward cleaner and

more sustainable energy systems under severe climatic conditions. Moreover, the results demonstrate a consistently lower LCOS for residential applications compared to office-building applications under climate-change conditions, both for centralized and distributed power supply. Quantitatively, the LCOS of PV-battery applications in residential buildings ranges from \$0.0527 to \$0.0555 kWh^{-1} , while the range for office buildings is from \$0.0597 to \$0.0633 kWh^{-1} . This disparity is because, compared to office buildings (average DoD during the 25-year simulation period at 0.2569), the operational charging/discharging DoD of battery in residential buildings (average DoD during the 25-year simulation period at 0.4126) is much higher due to the higher mismatching level between PV generations and building demands, leading to a slower lithium-ion battery-degradation ratio in the residential buildings. For the detailed battery-degradation mechanisms, refer to [Note S1](#) and [Figures S3](#) and [S4](#). The lower LCOS for battery applications in residential buildings suggests that homeowners, investors, and policy makers can prioritize battery integration in residential buildings due to the higher economic competitiveness and lower economic burden on public finance than battery integration in office buildings. Additionally, it is worth noting that the LCOS for the PV-battery system, ranging from \$0.0527 to \$0.0633 kWh^{-1} , is consistently lower than the levelized electricity price, which ranges from \$0.131 to \$0.134 kWh^{-1} for office buildings and \$0.066 to \$0.067 kWh^{-1} for residential buildings.³² This highlights the economic competitiveness of energy-storage technologies compared to grid-supplied power, demonstrating the necessity with economic feasibility of PV-battery systems to gradually replace the carbon-intensive grid power supply.

In order to adapt to various climate-change conditions, [Figure 6](#) compares system performance across four scenarios for

Figure 5. LCOS for four scenarios

Comparisons of LCOS in four scenarios under historical typical year, and RCP 2.6/RCP 4.5/RCP 8.5 for 2020/2030/2060/2100: scenario A, centralized PV-battery power system in office building; scenario B, distributed PV-battery power system in office building; scenario C, centralized PV-battery power system in residential building; scenario D, distributed PV-battery power system in residential building.

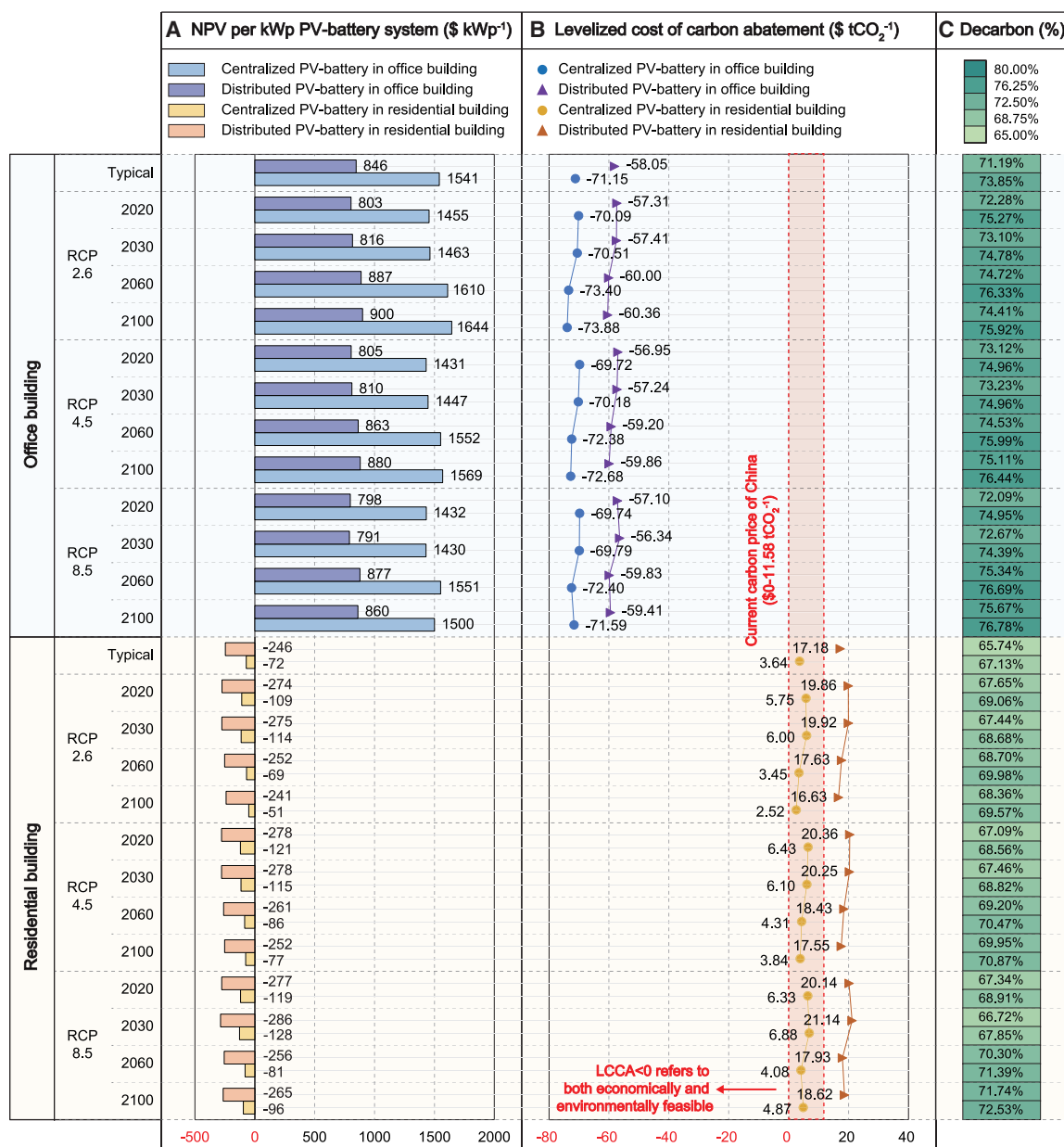


Figure 6. Economic-environmental performance of the integrative PV-battery-building system across four scenarios

Performance comparisons under historical typical year, RCP 2.6, RCP 4.5, and RCP 8.5: (A) NPV per kWp of PV-battery system; (B) LCCA; (C) decarbonization potential. Scenario A, centralized PV-battery power supplied for office building; scenario B, distributed PV-battery power supplied for office building; scenario C, centralized PV-battery power supplied for residential building; scenario D, distributed PV-battery power supplied for residential building.

both centralized and distributed PV-battery energy systems from both economic and ecological perspectives.

As shown in Figure 6, it is worth emphasizing that, across all RCPs, the NPV for all four scenarios exhibits an increasing trend from 2020 to 2100, which is also closely related to the increasing PV generation capacity influenced by climate change over time. This result suggests the financial viability and profitability of the ZEB project over the long term, underscoring its potential as a sustainable investment in the context of climate change. Moreover, it can be noted that the NPV per kWp of the PV-battery system

is higher in the centralized PV-battery systems compared to the distributed applications. For example, in the office building, the NPV ranges from \$1,430 to \$1,644 kWp⁻¹ in the centralized PV-battery system, which is significantly higher than the \$791–\$900 kWp⁻¹ of the distributed PV-battery systems. This discrepancy is primarily due to the higher PV generation capacity in the centralized configurations than that of the distributed PV applications, leading to greater cost savings with the centralized PV-battery integration, as described in Figure 1. This finding indicates the higher economic returns of PV-battery system

installations in the centralized sectors compared to the distributed sectors over the lifetime period of renewables. Moreover, it can be noted that the NPV is strongly affected by different types of buildings. Specifically, the NPVs of PV-battery applications in the office buildings are positive, while the NPVs of PV-battery applications in the residential buildings are negative. This result highlights the economic unfeasibility of the PV-battery application in residential buildings over the life-cycle period. This considerable gap can be attributed to two key factors: (1) the lower matching level between BIPV/PV power generation and consumption patterns in the residential building (i.e., PV power supply at daytime does not match with energy consumptions at morning and night) than that in the office building (i.e., PV power supply at daytime matches well with energy consumption at daytime); (2) the lower electricity price of the residential building (\$0.066–\$0.067 kWh^{−1}) compared to that of the office building (\$0.131–\$0.134 kWh^{−1}),³² which reduces the cost savings from the PV-battery system in the residential building. In order to quantify the decarbonization cost, the LCCA of the PV-battery system is calculated. Under climate-change conditions, the LCCA decreases over time from 2020 to 2100 across all RCPs, highlighting the lifetime cost payment declines and high investment motivations in decarbonization through PV-battery systems. Moreover, the centralized PV-battery applications consistently exhibit lower LCCA values compared to the distributed applications. For instance, the LCCA ranges from \$16.63 to \$21.14 tCO₂^{−1} in the distributed power supply for residential buildings, while the LCCA is only \$2.52–\$6.88 tCO₂^{−1} in centralized power supply systems. Results demonstrate the superiority of the centralized PV-battery application over the distributed application, and local policy makers can more easily initiate the decarbonization policy with economic subsidies on centralized PV-battery applications with low economic pressures. Meanwhile, the negative LCCA values observed for office-building applications suggest that prioritizing PV-battery investment could additionally improve the government's economic income and effectively alleviate the government's economic burden in decarbonization. However, by comparing the LCCA with the current carbon price, which ranges from \$0 to \$11.58 tCO₂^{−1} in China, the required LCCA for distributed PV-battery systems for residential building applications is still high. This highlights the challenges in scaling up renewable energy systems through LCCA-based incentive mechanisms, and local government may need to pay prioritized policies or economic subsidies to decarbonization technologies for widespread promotions.

From an ecological perspective, as shown in Figure 6C, by implementing PV-battery systems, a considerable decarbonization ratio of over 65% can be noticed with an upward trend from 2020 to 2100 across all RCPs, emphasizing the environmental sustainability and long-term effectiveness of the ZEB-based PV-battery system from the ecological perspective.

Impact of climate change on provincial system performance

The previous section offers comprehensive ZEB design principles and performance quantifications for various PV-battery-building application scenarios under climate-change conditions, which can serve as a robust data foundation for province-level

building energy-system analysis. To provide a more comprehensive perspective on the interconnected regional dynamics influencing the energy landscape, this section scales up the economic-environmental performance analysis to the provincial level based on the regional building types and areas. Results can help identify opportunities for optimizing resource allocation, infrastructure development, and energy mix optimization and can also inform policy makers and decision makers in developing more effective, regionally tailored energy policies under varying climate-change conditions.

Figure 7 demonstrates the provincial performance of the ZEB system, highlighting the significant influence of climate change on system metrics from perspectives of NPV, LCCA, and total decarbonization potential. As illustrated in Figure 7A, the total NPV exhibits a robust upward trend over time under all RCPs, increasing by 40.72% from \$614.61 × 10⁸ in 2020 to \$864.89 × 10⁸ in 2100 under the RCP 2.6 scenario. Moreover, the average NPV per kWp of the PV-battery system reaches a remarkable \$160–\$179 kWp^{−1} by 2100 across the RCP 2.6, RCP 4.5, and RCP 8.5 pathways. Meanwhile, with the integrative PV-battery system, Figure 7B reveals a notable decline in the LCCA along with the time, which decreases from \$−8.70 tCO₂^{−1} in 2020 to \$−12.25 tCO₂^{−1} in 2100 under RCP 2.6, from \$−8.61 tCO₂^{−1} in 2020 to \$−11.58 tCO₂^{−1} in 2100 under RCP 4.5, and from \$−8.41 tCO₂^{−1} in 2020 to \$−11.20 tCO₂^{−1} in 2100 under RCP 8.5. Note that the negative values of LCCA indicate that the deployment of the PV-battery system under climate-change conditions not only enhances economic performance but also effectively reduces carbon emissions, referring to both economic-environmental feasibility of the ZEB-based PV-battery system.

From ecological aspects, as shown in Figure 7C, the decarbonization potentials of the installed PV-battery system in Beijing steadily increase as time goes by under different RCPs with climate change. Specifically, the total emissions from buildings decrease from 31.31 × 10⁸ tons in 2020 to 29.53 × 10⁸ tons in 2100 by 5.67% under RCP 2.6, from 31.59 × 10⁸ tons in 2020 to 27.81 × 10⁸ tons in 2100 by 11.98% under RCP 4.5, and from 32.05 × 10⁸ tons in 2020 to 25.37 × 10⁸ tons in 2100 by 20.84% under RCP 8.5. Notably, the total decarbonization ratio of the installed PV-battery system can reach as high as a remarkable 70%, which highlights the effectiveness of the developed ZEB solution in contributing to the decarbonization process, even under challenging climate-change conditions.

Overall, it can be noticed that, compared to the isolated PV system without energy storages, although the integration of battery brings higher initial investments and embodied carbon emissions, the battery integration delivers significant improvements in both economic and environmental performances, by improving renewable penetration and mitigating the system independence on carbon-intensive and high-price grid power. Specifically, the integration of batteries can increase the NPV or even reverse the negative NPV of standalone PV systems into the positive NPV, making PV-battery systems economically viable (Figure 7A). Meanwhile, the decarbonization potential can be increased by about 30% with battery integration (Figure 7C), and the improvement becomes increasingly pronounced over time under all RCPs. These findings highlight the far-reaching

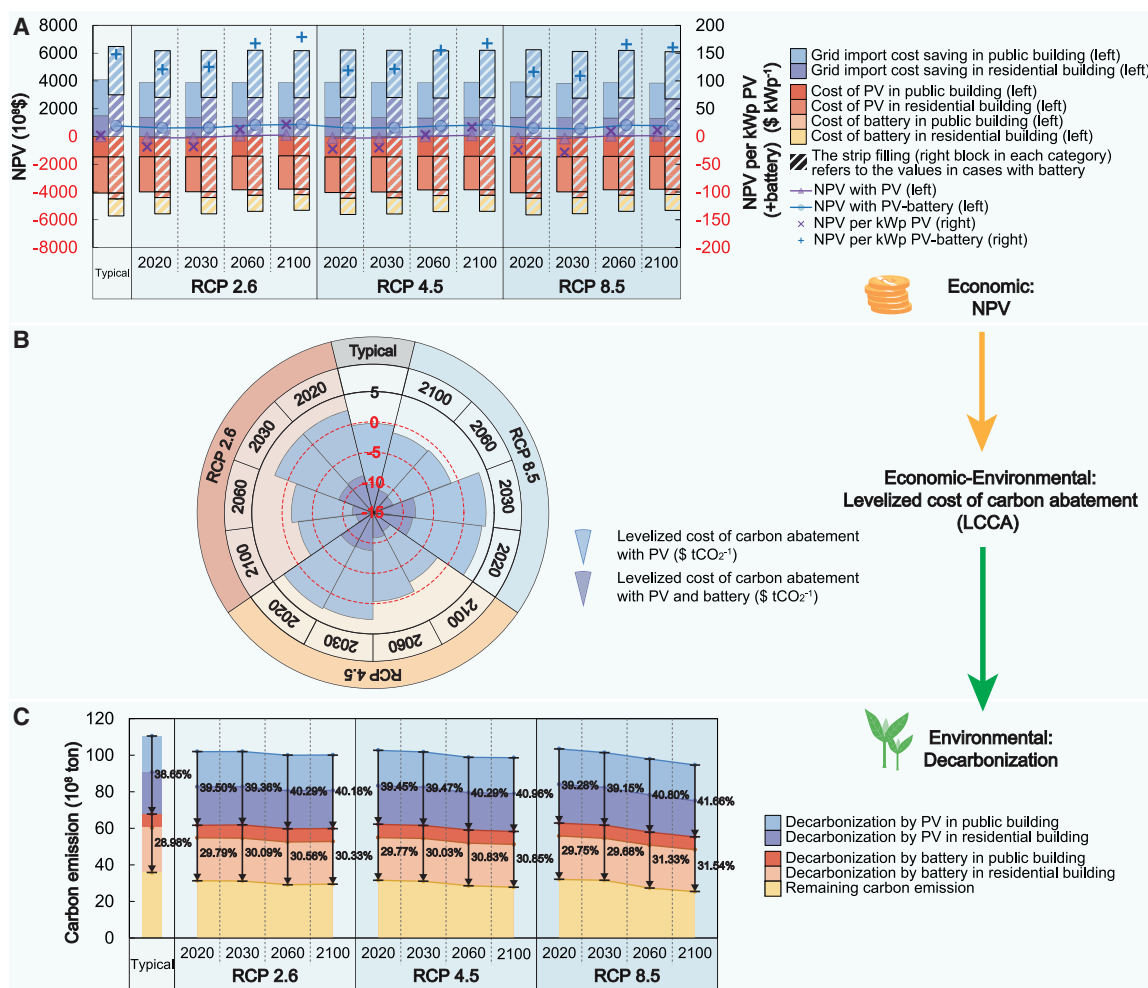


Figure 7. Economic-environmental performance of the integrative PV-battery-building system in Beijing
Comparisons on (A) NPV, (B) LCCA, (C) decarbonization potentials of buildings in Beijing under historical typical year, RCP 2.6, RCP 4.5, and RCP 8.5.

significance of integrating battery storage with PV systems for advancing the economic and environmental viability of clean-energy solutions under the elusive climate-change conditions, and demonstrating the superiorities of the developed PV-battery-building system based on the proposed kWp-kWh-m² design principles under evolving climate-change conditions.

Uncertainty and sensitivity analysis

Uncertainty can be classified into aleatory uncertainty and epistemic categories, arising from inherent randomness, natural variability, and information deficiencies. The identification of uncertainties can enhance decision making, risk identification, resource allocation, model validation, and regulatory compliance.³³ Within the PV-battery-integrated ZEB systems, the cost of PV, battery, and discount rate directly influence both the initial investments and operational costs, thereby affecting the system's overall economic viability and environmental performance throughout its lifespan. However, the values of these parameters are subject to fluctuations and unpredictability due

to evolving techniques, raw material availability, market dynamics, government policies, and global economic conditions. To comprehensively assess the performance of the proposed ZEB system under dynamic and future-proof conditions, this section conducts uncertainty and sensitivity analysis on the provincial LCCA, which reflects the economic and environmental competitiveness of the integrated PV-battery system, in relation to the key parameters, i.e., PV cost, battery cost, and discount rate. The analysis is carried out for Beijing, using the climate projection for the year 2100 under RCP 8.5.

As illustrated in Figure S8, the LCCA in Beijing steadily decreases with reductions in PV and battery costs, and demonstrates greater sensitivity to changes in PV costs compared to battery costs. Specifically, with the discount rate at 0%, the LCCA decreases by \$5.61 tCO₂⁻¹ for a 10% reduction in PV costs, while it decreases by only \$2.33 tCO₂⁻¹ for a 10% decrease in battery costs. This sensitivity is attributed to the higher proportion of PV costs relative to battery costs, which is approximately 2.5:1. Furthermore, it can be noticed that the LCCA significantly

increases with rising discount rate, shifting from a negative value of $\$-30.68 \text{ tCO}_2^{-1}$ at a 0% discount rate to a positive value of $\$27.55 \text{ tCO}_2^{-1}$ at a 10% discount rate (both under the current PV and battery costs). This deterioration in LCCA is due to the diminished cost savings from PV-battery integration over the life cycle at higher discount rates. In conclusion, the results indicate that reduced PV and battery costs can enhance the affordability and adoption of ZEB-based PV-battery systems, while the increasing discount rates may lead to high carbon abatement costs and even render the ZEB system economically unviable. This highlights the tension between economic considerations and environmental sustainability goals, suggesting an urgent need for policies and incentives that can bridge this gap to support long-term climate-change mitigation efforts.

Scalability and climate-change resilience in China

The impact of climate change on the transitions toward sustainable building energy systems is profoundly shaped by factors such as local climatic conditions, resource availability, grid energy structure, building types and areas, and electricity prices. To comprehensively capture the energy-system variations across broader geographic areas under climate-change conditions, this section scales up the energy-system design and performance analysis from Beijing to five representative provinces with distinct climate patterns in China. This scalability enables the holistic understanding of climate change's impact on regional energy-system dynamics, ultimately informing policy makers on the determination of effective and transferable strategies to prompt sustainable transitions across diverse climatic regions.

Figure 8 compares the required PV and battery capacities for ZEB systems across five distinct climatic regions in China under various climate-change conditions. Note that the weather condition in each province is represented by its provincial capital city, i.e., Xinjiang (Urumqi), Yunnan (Kunming), and Guangdong (Guangzhou). Beijing and Shanghai are municipalities. The results reveal significant regional variations in the PV-battery capacity requirements under climate-change conditions. Notably, the results reveal that, under the RCP scenarios, as the time period transits from 2020 to 2100, the harsh climate zones (like severe cold zone Xinjiang [Urumqi], cold zone Beijing, hot summer and cold winter zone [Shanghai]) exhibit a decreasing trend in the required PV and battery capacity, while the moderate climate zones (like temperate zone-Yunnan [Kunming] and hot summer and warm winter zone-Guangdong [Guangzhou]) show an increasing trend in the required PV and battery capacity. This is because future climate change will increase building electricity demands in cooling-dominated regions (like temperate zone Yunnan [Kunming] and hot summer and warm winter zone Guangdong [Guangzhou]) but reduce building electricity demands in heating-dominated regions (like severe cold zone Xinjiang [Urumqi], cold zone Beijing, and hot summer and cold winter zone Shanghai) (for details, refer to Figures S1 and S2). These findings highlight the regional variations in climate-change adaptation and the potential of ZEB solutions, informing guidelines for targeted investments in renewable-storage deployment and the formulation of tailored policy interventions across different geographic regions. Addi-

tionally, the results reveal that the harsh climate zones require a larger relative battery capacity compared to moderate climate zones. For example, the average ratio of required battery capacity to PV capacity ranges from 2.61 to 2.77 kWh kWp⁻¹ in Xinjiang (severe cold zone), while it is only 2.13–2.32 kWh kWp⁻¹ in Guangdong (hot summer warm and winter zone) (for the detailed optimal battery capacity, refer to Figure S6). This result can guide and optimize the PV-battery system design for regions with varying climatic conditions.

Figure 9 compares the system performance from both economic and ecological perspectives across five climatic regions under different climate-change conditions. From life-cycle economic feasibility analysis, the NPV per kWp of the PV-battery system represents the average value of return on investment of the developed PV-battery system following the ZEB principles and the proposed kWp-kWh-m² design framework. For temporal analysis under climate-change conditions, the results reveal an increasing trend in the NPV per kWp of the PV-battery system from 2020 to 2100 under all RCP scenarios, except for Xinjiang with time transitions from 2060 to 2100 under RCP 8.5. The increases are attributed to the improving PV generation capacity driven by the rise in solar radiation under climate-change conditions, while the decrease in Xinjiang under RCP 8.5 is caused by (1) PV production reduction due to substantial temperature rises in decreasing the PV efficiency, which overwhelms the positive impacts of increased solar radiation under severe climatic conditions; (2) significant demand reduction as well as the low local electricity price, which limits the cost-saving benefits of PV-battery integration; (3) the large battery requirement in Xinjiang, which increases the initial investments of the PV-battery system. However, the general increasing trend of NPV across the five regions indicates the long-term investment feasibility under climate-change conditions. Meanwhile, it should be emphasized that the NPV per kWp is closely related to the local PV production capacity and electricity prices. For the PV generations of the climatic regions under climate-change conditions, refer to Figure S5. For the detailed electricity prices, refer to Figure S7 and Tables S9 and S10. Specifically, regions with higher electricity prices and higher PV production capacity lead to a higher NPV. For example, although Beijing and Shanghai show similar average electricity price levels ($\$0.0713$ – $\$0.0741 \text{ kWh}^{-1}$ for Beijing and $\$0.0722$ – $\$0.0748 \text{ kWh}^{-1}$ for Shanghai), Beijing with higher PV productions (annual production ranges at 1,327.27–1,426.53 kWh kWp⁻¹ of Beijing compared to 1,093.66–1,267.31 kWh kWp⁻¹ for Shanghai) shows consistently higher NPV per kWp of PV-battery installations at $\$109.22$ – $\$179.15 \text{ kWp}^{-1}$ compared to $\$19.00$ – $\$161.86 \text{ kWp}^{-1}$ for Shanghai. Meanwhile, compared to Beijing (annual production at 1,327.27–1,426.53 kWh kWp⁻¹ and electricity price at $\$0.0713$ – $\$0.0741 \text{ kWh}^{-1}$), Xinjiang with similar PV production capacities (annual production ranges at 1,337.69–1,499.78 kWh kWp⁻¹) but lower electricity prices ($\$0.0435$ – $\$0.0438 \text{ kWh}^{-1}$) shows a lower NPV ($\-405.26 to $\$-373.14 \text{ kWp}^{-1}$).

In terms of the environmental competitiveness of the ZEB system, the decarbonization potentials of the integrated PV-battery systems improve as time moves from 2020 to 2100 under all RCPs, and this increase is more obvious in regions with harsher climates (e.g., in Xinjiang [Urumqi] severe cold zone, the decarbonization potential rises from 60.93% in 2020 to 64.32% in

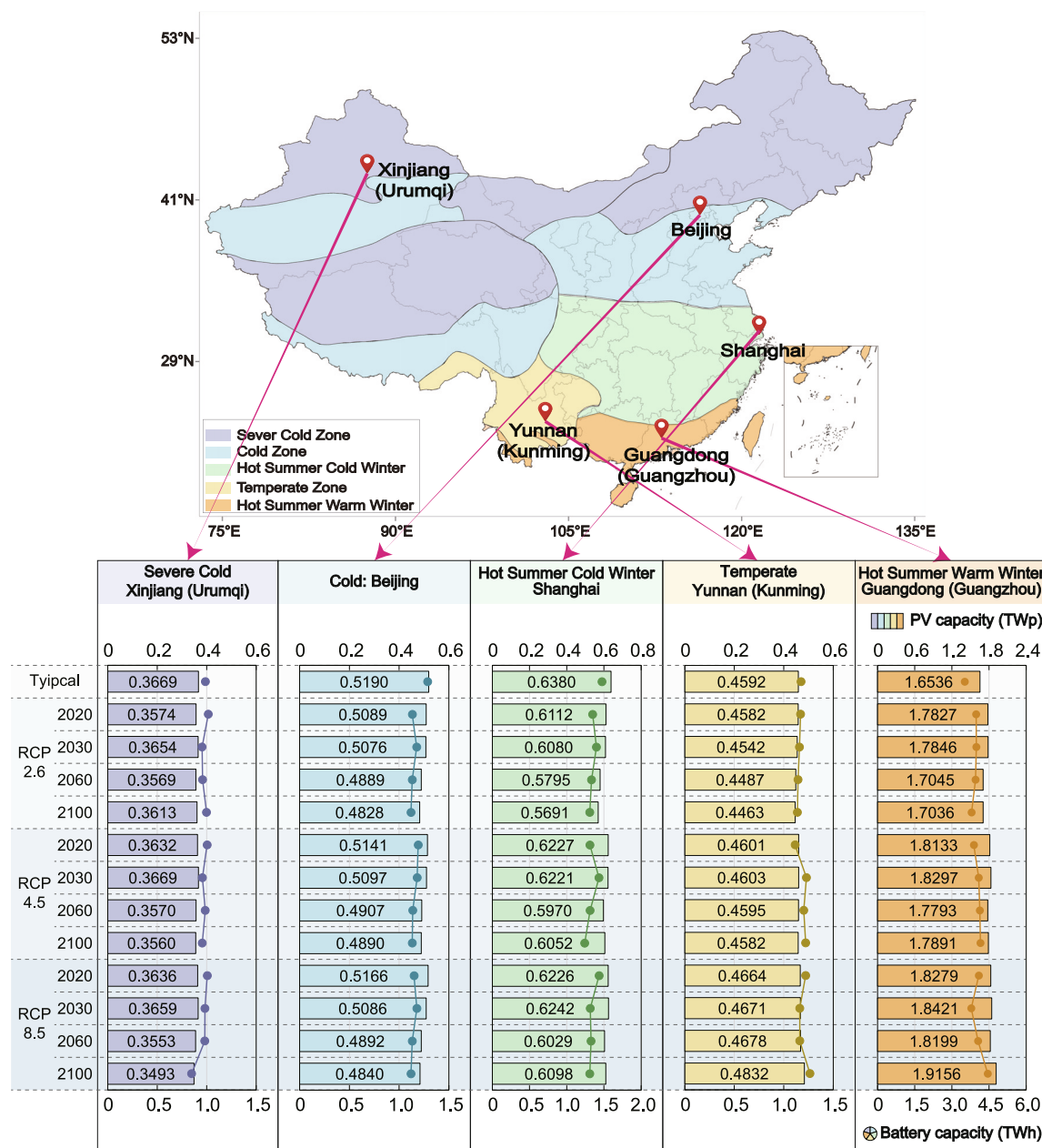


Figure 8. Required PV-battery capacity for distinct climatic regions

Comparisons of the total required PV-battery capacity for ZEB transitions following the optimal PV-battery-building system design principles (kWp-kWh-m^2) across five climate regions in China under climate-change conditions. Building demand, PV generation, and optimal battery capacity of the five climatic regions under climate-change conditions refer to Figures S1–S2, S5, and S6, respectively. For building types and areas of the five regions, refer to Table S8.

2100 under RCP 8.5). This improvement is primarily due to the increase in PV production capacity and the key role of battery in supporting energy flexibility and resilience under varying climate-change conditions. Furthermore, the decarbonization potential of the PV-battery integration is significantly affected by the energy structure of the local grid. The results reveal that the PV-battery integration achieves more significant carbon reductions in regions with a higher grid carbon emission factor (CEF) (for the CEF calculation model and results, refer to Note

S2 and Tables S11 and S12). For instance, compared to Beijing with a grid CEF of $0.8882 \text{ kg kWh}^{-1}$ and decarbonization potentials at 67.63%–73.20%, the PV-battery system in Yunnan with a lower grid CEF of $0.0958 \text{ kg kWh}^{-1}$ shows lower decarbonization potentials at 41.05%–44.30%. This is primarily due to the high proportion of hydropower with 82.4% of the grid energy structure in Yunnan, resulting in a low grid CEF, and ultimately reducing the environmental superiority of renewable-supplied power over grid supply electricity.

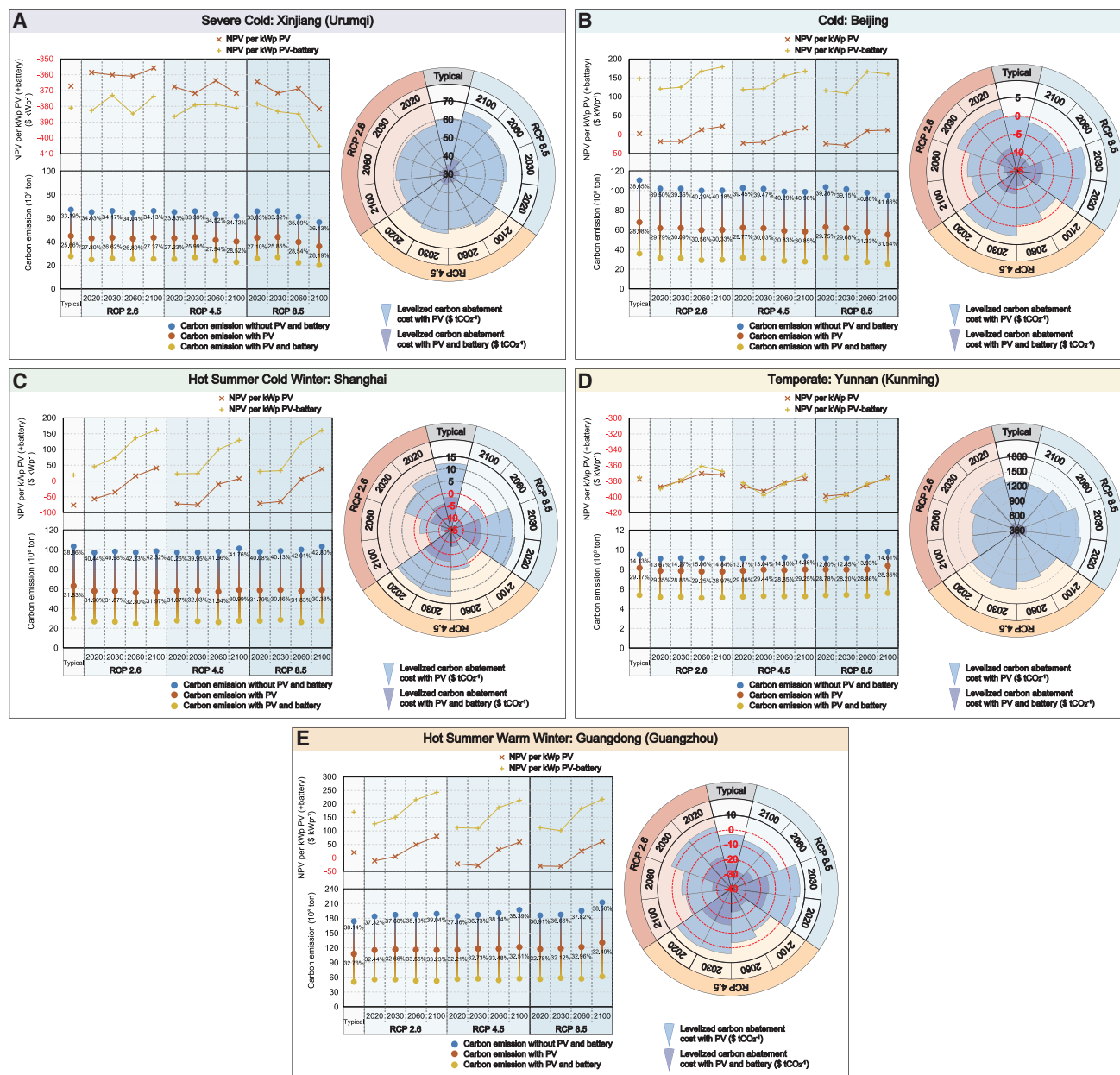


Figure 9. Economic-environmental performance of the integrative PV-battery-building system across distinct climatic regions

Comparisons of NPV per kWp of the PV-battery system, LCCA, and decarbonization potentials across five climate regions in China under climate-change conditions. (A) severe cold region: Xinjiang; (B) cold region: Beijing; (C) hot summer cold winter region: Shanghai; (D) temperate region: Yunnan; (E) hot summer warm winter region: Guangdong.

Regarding the LCCA, which is determined by both NPV and decarbonization, it is notable that the LCCA differs greatly by regions. Specifically, LCCA ranges from $-\$30.26 \text{ tCO}_2^{-1}$ to $\$8.31 \text{ tCO}_2^{-1}$ in Guangdong, from $-\$12.96 \text{ tCO}_2^{-1}$ to $\$12.16 \text{ tCO}_2^{-1}$ in Shanghai, and from $-\$12.25 \text{ tCO}_2^{-1}$ to $\$3.66 \text{ tCO}_2^{-1}$ in Beijing, while high LCCA values in Yunnan and Xinjiang can be noticed, i.e., from $\$398.75 \text{ tCO}_2^{-1}$ to $\$1,613.47 \text{ tCO}_2^{-1}$ in Yunnan and from $\$33.17 \text{ tCO}_2^{-1}$ to $\$65.22 \text{ tCO}_2^{-1}$ in Xinjiang. By comparing the LCCA with the current carbon price ranging from $\$0$ to $\$11.58 \text{ tCO}_2^{-1}$ in China, the proposed PV-battery capacity design

approach for ZEB transitions in Guangdong, Shanghai, and Beijing demonstrates the economic feasibility from a policy perspective, while it still remains challenging in Yunnan and Xinjiang due to the considerably high LCCA. In other words, the proposed PV-battery capacity design approach is applicable for ZEB transitions in Guangdong, Shanghai, and Beijing, while additional economic subsidy is necessary to promote the PV-battery decarbonization technologies in Yunnan and Xinjiang.

Last but not least, the significant contribution of batteries in improving regional economic-environmental performance under

climate-change conditions can be consistently observed across the five distinct climate regions. Quantitatively, compared to the isolated PV building systems without energy storage, the integration of batteries can further contribute to the building decarbonization by 25.66%–42.80% and reduce the required LCCA by 40.36%–69.10% or even reverse the positive LCCA to negative values. This ensures the spontaneous PV-battery installation without the necessity of economic subsidy or policy support.

The above comparative analysis highlights the variations of the climate-change impact across the five regions considering distinctive renewable sources, climatic patterns, grid energy structure, building types and areas, and electricity prices. These results can guide the prioritization of regional ZEB investments and provide long-term economic-environment policy-viability analysis for the ZEB system under climate-change conditions.

Global policies in renewable system promotion

During the sustainability transitions under severe climate change, governments have implemented a range of policies to facilitate renewable integration, including direct subsidies, tax incentives, carbon pricing, green certificates, and renewable energy targets.^{34,35} These measures have proved to be effective in encouraging investment, promoting innovation, setting targets, and fostering market establishment. Among these policies, the feed-in tariff (FIT) has been shown to be the most efficient and cost-effective way to foster renewable energy, as it provides direct payments to renewable energy providers while enabling the integration of clean energy into the grid.³⁶ This section analyzes the potential of FITs of different global regions in promoting the proposed ZEB-based PV-battery system, assessing the feasibility of this policy in China. The analysis is carried out based on the historical typical climate condition.

The literature review reveals significant disparities in FITs across regions, ranging from \$0 to \$0.59 kWh^{−1}.^{37–44} For this analysis, we select the following FIT values for a comprehensive range analysis: 0, \$0.007 kWh^{−1} (mean of Southeast Asia³⁷), \$0.14 kWh^{−1} (California, USA⁴⁰), \$0.23 kWh^{−1} (maximum of Southeast Asia³⁷), \$0.32 kWh^{−1} (Hong Kong, China⁴³), and \$0.59 kWh^{−1} (Canada⁴⁴). These values will be used to assess the impact of varying levels of FITs on economic and environmental improvements across distinct climatic regions in China, as indicated by the NPV and the LCCA.

According to Figure 10, the results demonstrate the significant impact of FITs on enhancing economic and ecological performance, potentially transforming a system from economic unviability to viability. Quantitatively, as the FIT increases from \$0 to \$0.59 kWh^{−1} in Beijing, the total NPV of the proposed PV-battery system rises from 763.73×10^8 to $15,112.45 \times 10^8$, with the LCCA decreasing from $-10.27 \text{ tCO}_2^{-1}$ to $-202.19 \text{ tCO}_2^{-1}$, indicating greater economic feasibility compared to the scenario without FIT policy. Moreover, in Yunnan, compared to the scenario without FIT, the NPV shifts from a negative value of $-1,724.74 \times 10^8$ to the positive value of $5,034.81 \times 10^8$, with LCCA reversing from $419.41 \text{ tCO}_2^{-1}$ to $-1,224.32 \text{ tCO}_2^{-1}$, demonstrating a transformation from being economically unfeasible to being both economically and environmentally viable. These results underscore the effectiveness of FITs in promoting spontaneous installations by incentivizing system owners.

However, it should be noted that the effectiveness of FITs varies significantly by region. For example, compared to the scenario without FIT, the FIT of \$0.59 kWh^{−1} reverses the LCCA in Yunnan from $419.41 \text{ tCO}_2^{-1}$ to $-1,224.32 \text{ tCO}_2^{-1}$, by $1,643.73 \text{ tCO}_2^{-1}$, but only reverses the LCCA in Xinjiang from 35.34 tCO_2^{-1} to $-372.41 \text{ tCO}_2^{-1}$, by $407.75 \text{ tCO}_2^{-1}$. This disparity is primarily due to the lower CEF of the grid in Yunnan ($0.0958 \text{ kg kWh}^{-1}$ due to a high proportion of hydropower at 82.4%, compared to $0.7164 \text{ kg kWh}^{-1}$ in Xinjiang). The cleaner grid-energy structure in Yunnan results in limited CO₂ emissions that can be mitigated by the PV-battery system, making the carbon abatement cost highly sensitive to economic benefits. This finding highlights the necessity of ensuring economic benefits in regions with cleaner grid-energy structures to achieve further decarbonization goals. Overall, it can be concluded that the FITs should be carefully tailored to suit different regions; while high policy incentives effectively promote renewable installations, they may also impose significant economic burdens on local governments.

DISCUSSION

The unprecedented climate-change and global energy-shortage crisis urgently call for adaptive solutions to drive energy-resilient and sustainable transitions. In this study, a robust framework of integrative ZEB energy systems for bolstering climate-adaptive resilience has been proposed, with quantifiable strategic investment recommendations and targeted policy intervention guidance for distinct climatic regions toward sustainability transformations, addressing the challenges posed by evolving climate-change scenarios. A tailored kWp-kWh-m² design approach with intrinsic relationships of prosumer storage is applied, following the ZEB design principles and innovative U-value/M-value battery-sizing methods, enabling high renewable penetrations while maintaining balanced flexibility-cost-environmental impacts for both centralized and distributed energy systems. Comprehensive analysis from life-cycle techno-economic-environmental feasibility has been conducted, encompassing diverse application scenarios, dynamic equipment degradation, and varying climatic change conditions, providing rigorous evaluations for benefit-oriented investors or green development-targeted policy makers. Afterward, the proposed framework and evaluations are generalized and scaled to five climatic regions across China with distinct renewable resources, electricity consumption patterns, grid energy structures, building stocks, and electricity prices. This scalability enables a holistic understanding of the energy-system variations across broader geographic areas under climate-change conditions, contributing to identifying transferable strategies to drive sustainable transitions across diverse climatic regions. The insights and strategies derived from this work can serve as a valuable reference and generic approach for similar efforts in other regions facing the challenges of climate change and the pursuit of sustainable energy systems. Key results are summarized below.

- (1) Depending on climatic variability and electricity consumption patterns, future climate change can significantly increase the building electricity consumption in cooling-dominated regions (like Yunnan and Guangdong), but

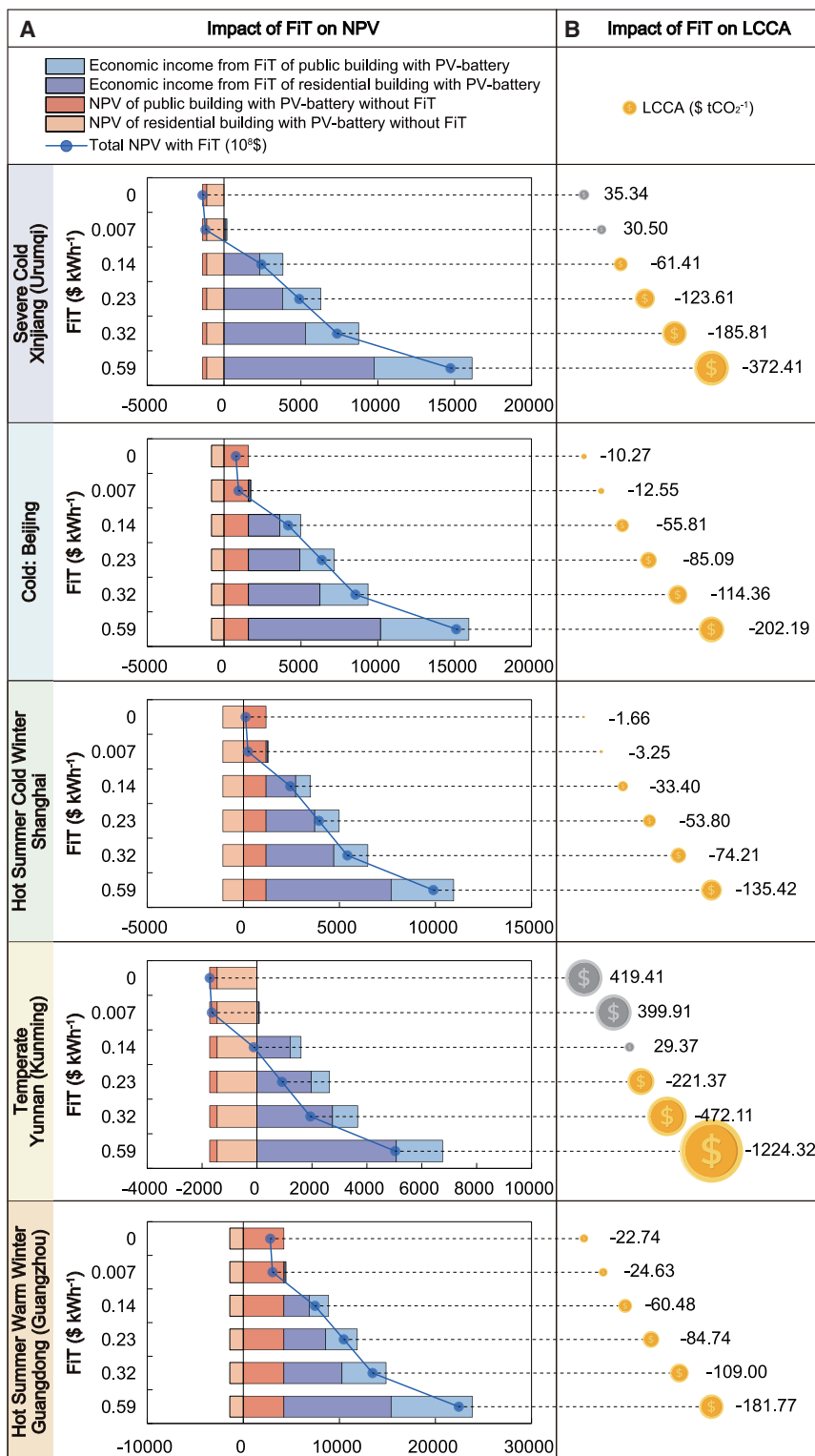


Figure 10. Impact of different FITs on NPV and LCCA improvement across five climate regions in China under historical typical climate

Negative values of the LCCA indicate that the installation of the proposed integrative PV-battery system not only effectively reduces carbon emissions but also provides economic benefits. (A) Impact of FIT on NPV; (B) Impact of FIT on LCCA.

- contrarily reduce the building electricity consumption in heating-dominated regions (like Xinjiang, Beijing, and Shanghai). For instance, the annual electricity demand of the office building located in Guangdong rises from 137.60 kWh m⁻² in 2020 to 158.42 kWh m⁻² in 2100 by 15.13% under the RCP 8.5 scenario. In contrast, in Xinjiang, the annual electricity demand of the residential building decreases from 219.29 kWh m⁻² in 2020 to 185.21 kWh m⁻² in 2100 by 15.54% under RCP 8.5.
- (2) The tailored kWp-kWh-m² design approach in this study can guide the design of integrative PV-building-battery systems in both centralized and distributed configurations. By applying the proposed design approach, centralized PV-supply systems require substantially larger battery capacity (e.g., 3.75–4 kWh kWp⁻¹ in Beijing) compared to distributed power supply systems (e.g., 1.75–2.75 kWh kWp⁻¹ in Beijing). Furthermore, office buildings with a higher matching between PV power supply and electricity demand exhibit smaller battery capacity requirements (e.g., 1.75–2 kWh kWp⁻¹ in Beijing) than that of residential buildings (e.g., 2.5–2.75 kWh kWp⁻¹ in Beijing), in which load patterns do not match well with PV generation. Regarding climatic factors, the harsh climate zones, such as Xinjiang (Urumqi) of the severe cold zone, necessitate a larger battery capacity for PV systems (e.g., 2.61–2.77 kWh kWp⁻¹) compared to more moderate climate zones, such as Guangdong (Guangzhou) with hot summer and warm winter climate (e.g., 2.13–2.32 kWh kWp⁻¹). Moreover, fluctuations in power supply and demand within ZEBs, driven by varying climate-change conditions, lead to differences in optimal system configurations. For instance, in Beijing under the RCP 2.6 scenario, the required PV and battery capacities per unit area decrease from 0.1490 kWp-0.5586 kWh-m² in 2020 to 0.1375 Wp-0.5156 kWh-m² in 2100 under RCP 2.6 within centralized PV-battery-supplied residential building systems. The results are invaluable in guiding rigorous system design for ZEB transitions across diverse applications and variable climatic conditions.
 - (3) Zero-carbon transformation through integrative PV-battery requires economic investment, while the LCCA is highly dependent on climate type, climate change, and building type. By generalizing and scaling the ZEB framework and the life-cycle economic-ecological performance evaluation to five distinct climatic regions in China, the LCCA demonstrates negative values in regions with abundant renewable resources, high electricity prices, and carbon-intensive grid structure, but it exhibits super-high values in less favorable regions. Specifically, the LCCA ranges from [\$-30.26, \$8.31] tCO₂⁻¹ in Guangdong to [\$398.75, \$1,613.47] tCO₂⁻¹ in Yunnan. Compared to the current carbon price ranging from \$0–\$11.58 tCO₂⁻¹ in China, the results highlight the diverse economic-environmental competitiveness and policy feasibility of the PV-battery within the ZEB context from policy perspectives, indicating regional strategic investment for zero-carbon transformations under climate-change conditions.

- (4) By avoiding the battery oversizing (leading to battery resource waste) and undersizing (resulting in low renewable penetration), the optimal battery integration following the proposed kWp-kWh-m² design framework is crucial for improving the life cycle economic-environmental feasibility of the ZEB system by achieving deep decarbonization and reducing the required LCCA. According to the results, compared to isolated PV systems, the battery integration under the proposed kWp-kWh-m² design approach can contribute to the decarbonization by 25.66%–42.80% and reduce the required LCCA by 40.36%–69.10%, and these positive effects become increasingly pronounced over time even under challenging climate-change conditions. These findings underscore the effectiveness of battery storage as a critical enabler for sustainable carbon neutrality transformations under climate change and emphasize the environmental sustainability and long-term economic feasibility of the proposed ZEB-based PV-battery system.

The research explores the climate-change resilience for ZEB transitions through a comprehensive micro-to-macro approach from supply-demand fluctuation, optimal ZEB system design principles, community-level system performance evaluation, to system scaling up and provincial configurations, encompassing diverse application scenarios, dynamic equipment degradation, life cycle assessment, and varying climatic conditions. The research results can provide invaluable guidelines on ZEB system planning, economic-environmental feasibility evaluation, and regional strategic renewable investment under climate-change conditions. However, the study acknowledges several limitations: (1) the work lacks the analysis of system economic-environmental factors, such as the evolution of equipment prices, electricity prices, and grid CEFs under climate-change conditions while focusing primarily on weather condition variations; (2) there is a lack of in-depth analysis of renewable types, such as wind turbine, hydropower, biomass, and nuclear power, neglecting the performance implications of mixed-renewable energy systems; (3) the research primarily focuses on first-hand battery integration in building systems, overlooking the wider-utilization and lower-cost potential of second-hand batteries retired from electric vehicles. Therefore, future work will focus on (1) incorporating comprehensive system evaluation with reliable predictions on economic-ecological parameter evolutions under different RCP scenarios; (2) conducting comprehensive analysis of mixed-renewable energy systems under climate-change conditions; (3) exploring the performance implications of integrating second-hand batteries into the ZEB system design; and (4) considering the complex sociotechnical uncertainties on renewable-battery-building systems under varying climate-change and economic conditions by applying advanced technologies such as expert-judgment-integrated Bayesian approaches and fuzzy-logic methods.⁴⁵

EXPERIMENTAL PROCEDURES

Simulation platform

The energy-based models are developed in TRNSYS 18,⁴⁶ a dynamic simulation software that excels in modeling complex systems including building

energy systems, renewable energy systems, and storage systems. TRNSYS enables precise and flexible system design and modeling, providing an effective platform for developing accurate and detailed energy-based models. The dynamic power flow for the integrative PV-battery building energy system is shown in Figure S9 and Note S3. For the building information and simulation environment, refer to Note S4.

U-value method for centralized battery sizing

The U-value method is employed to determine the optimal battery capacity in the centralized PV system, which is developed by Peters et al.¹³ In renewable-supplied scenarios, the U-value method identifies the optimal battery to improve the overlap between the unstable renewable supply and stochastic demand. A constant load is chosen because it presents the most challenging aspect for effective PV utilization across diverse load profiles.¹³

The integration of batteries in the renewable-supplied system enables the shifting of surplus renewable generation to demand shortage period, thereby mitigating the variability of renewable-battery power and smoothing out fluctuations. To quantify the effect, the uniform factor U, is introduced:

$$P_{\text{battery}}(t) = \begin{cases} \text{Min}\left(P_{\text{PV}}(t) - P_{\text{max}}(t), \left(\text{Cap}_{\text{battery}}(t) - \int P_{\text{battery}}(t)dt\right) / 90\%\right), P_{\text{PV}}(t) > P_{\text{max}}(t) \\ -\text{Min}\left(P_{\text{max}}(t) - P_{\text{PV}}(t), \int P_{\text{battery}}(t)dt\right), P_{\text{PV}}(t) < P_{\text{max}}(t) \end{cases} \quad \text{Equation (5)}$$

$$U = \frac{\int (P_{\text{PV}}(t) + P_{\text{battery}}(t) - P_{\text{curtailment}}(t))dt}{P_{\text{max}} \cdot \Delta t} \quad \text{Equation (1)}$$

where $P_{\text{PV}}(t)$ refers to the instantaneous PV generation. $P_{\text{curtailment}}(t)$ refers to the curtailed power at the time t . Δt refers to the simulation period (1 year in this study). $P_{\text{battery}}(t)$ is the power charged/discharged to/from the battery, mathematically given by:

$$P_{\text{battery}}(t) = \begin{cases} \text{Min}\left(P_{\text{PV}}(t) - P_{\text{max}}, \left(\text{Cap}_{\text{battery}}(t) - \int P_{\text{battery}}(t)dt\right) / 90\%\right), P_{\text{PV}}(t) > P_{\text{max}} \\ -\text{Min}\left(P_{\text{max}} - P_{\text{PV}}(t), \int P_{\text{battery}}(t)dt\right), P_{\text{PV}}(t) < P_{\text{max}} \end{cases} \quad \text{Equation (2)}$$

where P_{max} is the constant power load below which the PV power will not be curtailed. $\text{Cap}_{\text{battery}}(t)$ refers to the battery capacity considering the battery degradation during the charging/discharging cycles. 90% is the charging efficiency of the battery. P_{max} is derived by self-consistently solving the following equation:

$$c \cdot \int P_{\text{PV}}(t)dt = \int |P_{\text{PV}}(t) + P_{\text{battery}}(t) - P_{\text{max}}|dt \quad \text{Equation (3)}$$

Allowing a small amount of curtailment is justified to enhance both economic viability and grid stability. Here, c is the curtailment fraction of the total PV power generation, which is set to be 5%. Figure S10A elucidates the definition of

U value. As larger battery capacity is introduced, the uniformity of the PV-battery power output is enhanced, thereby improving the overlap between the PV-battery power profile and the load (P_{max}).

M-value method for distributed battery sizing

For battery integration in the distributed PV application scenario, the fluctuating load profile should be considered. Based on the U-value method, the M-value method is proposed to determine the optimal battery capacity in distributed applications. This approach aims to enhance the alignment between power supply and load profiles, focusing on their shapes and trends. To quantify the effect, the matching factor M, is introduced:

$$M = \frac{\int (P_{\text{PV}}(t) + P_{\text{battery}}(t) - P_{\text{curtailment}}(t))dt}{\int P_{\text{max}}(t)dt} \quad \text{Equation (4)}$$

where $P_{\text{PV}}(t)$ refers to the instantaneous PV generation. $P_{\text{curtailment}}(t)$ refers to the curtailed power at the time t , Δt refers to the simulation period (1 year in this study), and $P_{\text{battery}}(t)$ is the power charged/discharged to/from the battery, mathematically given by:

90% is the charging efficiency of the battery. $\text{Cap}_{\text{battery}}(t)$ refers to the battery capacity considering the battery degradation during the charging/discharging cycles.

The key difference between the U-value method and the M-value method is the definition of $P_{\text{max}}(t)$. For M value in the distributed application, $P_{\text{max}}(t)$ is defined as follows:

$$P_{\text{max}}(t) = r \cdot P_{\text{ED}}(t) \quad \text{Equation (6)}$$

$P_{\text{ED}}(t)$ is the electricity demand of the investigated scenario at time t . $P_{\text{max}}(t)$ is defined as the ratio of the electricity demand, where the ratio r always greater than 1. In that way, the PV-battery power curves are approaching the $P_{\text{max}}(t)$ curve with larger battery integration, representing the improvement of the alignment between power supply and load profiles in shapes and trends.

The ratio r is derived by self-consistently solving the following equation:

$$c \cdot \int P_{\text{PV}}(t)dt = \int |P_{\text{PV}}(t) + P_{\text{battery}}(t) - r \cdot P_{\text{ED}}(t)|dt \quad \text{Equation (7)}$$

Likewise, c is the curtailment fraction of the total PV power generation, which is also set to be 5%. Figure S10B illustrates the definition of the M value. It can be seen that, with larger battery integration, the PV-battery output and load curves gradually align in shape, representing the improvement in renewable penetration and power supply reliability.

The upper part of Figure S10C shows the uniformity factor (matching factor) as a function of battery capacity. The lower part of Figure S10C demonstrates the first derivative of these functions. The optimal battery capacity is determined by the point with the highest increase in the uniformity or matching factor, which aims to maximize the marginal efficiency of battery integration, ensuring the greatest improvement in load matching while achieving a compromise between flexibility-cost-environmental impacts.

To further highlight the superiorities of the developed methods, the comparative analysis with traditional battery-sizing methodologies is shown in Note S5 and Figure S11.

NPV calculation model

In this research, NPV refers to the 25-year return on investment of the integrative PV (and battery) system. For NPV of the isolated PV system, the calculation methods are as follows:

$$NPV_{PV} = \sum_{n=0}^N \frac{CS_{grid,PV,n}}{(1+r)^n} - InV_{PV} - \sum_{n=0}^N \frac{O\&M_{PV,n}}{(1+r)^n} \quad \text{Equation (8)}$$

where $CS_{grid,PV,n}$ refers to the cost saving of reduced grid imported electricity with PV integration in the n_{th} year. InV_{PV} is the initial investment of PV/BIPV systems, which is $\$715 \text{ kWp}^{-1}$. $O\&M_{PV,n}$ is the operation and maintenance cost of PV/BIPV system in the n_{th} year, which is $\$3.4 \text{ kWp}^{-1}$ per year. r is the discount rate at 2%. N is the lifetime of the investigated renewable energy devices, i.e., 25 years.⁴⁷

In terms of the PV-battery system, the NPV is calculated as follows:

$$NPV_{PV-Bat} = \sum_{n=0}^N \frac{CS_{grid,PV-Bat,n}}{(1+r)^n} - InV_{PV} - \sum_{n=0}^N \frac{O\&M_{PV,n}}{(1+r)^n} - InV_{Bat} - \sum_{n=0}^N \frac{O\&M_{Bat,n}}{(1+r)^n} \quad \text{Equation (9)}$$

where $CS_{grid,PV-Bat,n}$ is the cost saving of reduced grid-imported electricity with PV and battery integration in the n_{th} year. $\sum_{n=0}^N \frac{O\&M_{Bat,n}}{(1+r)^n}$ refers to the operation and maintenance cost of the integrated battery, which is assumed to be 1% of the initial cost of the integrated battery (InV_{Bat}).⁴⁹ For the initial investment of the integrated battery (InV_{Bat}), during the 25-year simulation period, more than one battery will be used and retired in some scenarios (for the battery degradation model, refer to Note S1, Figures S3 and S4, and Table S7). Therefore, the InV_{Bat} and operation and maintenance costs of the battery can be calculated as follows:

$$InV_{Bat} = P_{Battery \text{ pack}} \times Cap_{Bat} \times (N_{retired} + D_{Bat,N+1} / \Delta D_{Bat}) \quad \text{Equation (10)}$$

$$\sum_{n=0}^N \frac{O\&M_{Bat,n}}{(1+r)^n} = InV_{Bat} \times 1\% \quad \text{Equation (11)}$$

where $P_{battery \text{ pack}}$ refers to the battery pack price at $\$127 \text{ kWh}^{-1}$.⁵⁰ Cap_{Bat} refers to the battery capacity (kWh). $N_{retired}$ refers to the replacement times of batteries when they are retired with a degradation ratio greater than 40%⁴⁸ during the 25-year lifetime. $D_{Bat,N+1}$ refers to the magnitude of the $(N+1)_{th}$ battery degradation. ΔD_{Bat} is the magnitude of battery degradation before retirement at 40%. The equivalent number of battery replacements of the $(N+1)_{th}$ battery is identified as $D_{Bat,N+1} / \Delta D_{Bat}$.

LCOS calculation model

The LCOS refers to the life cycle cost of energy storage, considering the initial investment, operational and maintenance costs, and the lifetime of the storage technique. Here we investigate the LCOS that is independent of the charging cost, which is most broadly applicable.⁵¹ The calculation model is as follows:

$$LCOS = \frac{InV_{Bat} + \sum_{n=0}^N \frac{O\&M_{Bat,n}}{(1+r)^n}}{\sum_{n=0}^N \frac{E_{Bat,dischar,n}}{(1+r)^n}} \quad \text{Equation (12)}$$

where InV_{Bat} refers to the initial investment of the integrated battery (as calculated by Equation 10). $\sum_{n=0}^N \frac{O\&M_{Bat,n}}{(1+r)^n}$ refers to the total operation and maintenance cost of the battery system over the lifetime (as calculated by Equation 11). $E_{Bat,dischar,n}$ refers to the energy discharged from the battery system to supply the demand shortage in the n_{th} year.

Decarbonization calculation model

The decarbonization ratio is calculated based on the reference case without PV or battery integration to determine the decarbonization potential of the integrative PV (and battery) in building energy systems. For the isolated PV system, the calculation methods are as follows:

$$D_{PV} = D_{grid,PV} - EmC_{PV} \quad \text{Equation (13)}$$

$$D_{grid,PV} = CEF_{grid} \cdot \int_{n=0}^N P_{grid,imp \text{ reduced with PV}}(t) dt \quad \text{Equation (14)}$$

where EmC_{PV} refers to the embodied carbon of the PV devices, which considers the carbon emission during manufacturing, transportation, maintenance, and recycling of renewable energy equipment, and the carbon emission during the operational stage is considered to be 0.⁵² The EmC is 560 kg $\text{CO}_2 \text{ kWp}^{-1}$ for PV/BIPV.⁵³ $D_{grid,PV}$ is the decarbonization of reduced grid-imported electricity with PV integration, which is calculated by the carbon emission factor of the regional grid (CEF_{grid}) and the total reduced grid-imported energy ($\int_{n=0}^N P_{grid,imp \text{ reduced with PV}}(t) dt$). For the CEF calculation model and results, refer to Note S2 and Tables S11 and S12.

Likewise, the decarbonization potential of the PV-battery system is calculated by the embodied carbon of PV devices (EmC_{PV}), embodied carbon of battery (EmC_{Bat}), and the decarbonization of reduced grid imported electricity with PV-battery integration ($D_{grid,PV-Bat}$).

$$D_{PV-Bat} = D_{grid,PV-Bat} - EmC_{PV} - EmC_{Bat} \quad \text{Equation (15)}$$

$$D_{grid,PV-Bat} = CEF_{grid} \cdot \int_{n=0}^N P_{grid,imp \text{ reduced with PV-Bat}}(t) dt \quad \text{Equation (16)}$$

$$EmC_{Bat} = 42 \times Cap_{Bat} \times (N_{retired} + D_{Bat,N+1} / \Delta D_{Bat}) \quad \text{Equation (17)}$$

The embodied carbon of the lithium-ion battery is 42 kg kWh^{-1} .⁵⁴ $N_{retired}$ refers to the replacement times of batteries when they are retired with a degradation ratio greater than 40% during the 25-year lifetime. $D_{Bat,N+1}$ refers to the magnitude of the $(N+1)_{th}$ battery degradation. ΔD_{Bat} is the magnitude of battery degradation before retirement at 40%. The equivalent number of battery replacements of the $(N+1)_{th}$ battery is identified as $D_{Bat,N+1} / \Delta D_{Bat}$.

LCCA calculation model

LCCA (unit in $\$ \text{ tCO}_2^{-1}$) calculates the average cost of reducing 1 ton of carbon dioxide emissions over the lifetime of a project or technology, which represents the economic-environmental competitiveness of the investigated PV-battery system in building energy system. The calculation methods are as follows:

$$LCCA_{PV} = \frac{-NPV_{PV} \cdot 1000}{D_{PV}} \quad \text{Equation (18)}$$

$$LCCA_{PV-Bat} = \frac{-NPV_{PV-Bat} \cdot 1000}{D_{PV-Bat}} \quad \text{Equation (19)}$$

where NPV_{PV} (NPV_{PV-Bat}) (unit in $\$$) refers to the total NPV of the PV (PV-battery) integration, and D_{PV} (NPV_{PV-Bat}) (unit in kg CO_2) refers to the decarbonization ratio of the PV (PV-battery) integration over the lifetime periods.

Modeling approach

There were several assumptions: notably, the analysis deliberately disregards the complex economic and environmental factors, such as the evolution of equipment prices, electricity prices, and grid energy structure. This exclusion is rooted in the recognition that predictions regarding parameters with intricate, interacting factors become highly unreliable, particularly for long-term projections (e.g., 2060 and 2100). By grounding the discussion in credible weather data, the reliability of the research results can be ensured.

RESOURCE AVAILABILITY

Lead contact

Please contact the lead contact, Yue-Kuan Zhou (yuekuan.zhou@outlook.com; yuekuanzhou@hkust-gz.edu.cn) for information related to the data and code described in the following “Experimental procedures” section.

Materials availability

This study did not generate new unique materials.

Data and code availability

The authors declare that the data used as model inputs supporting the findings of this study are available within the paper and supplemental information. The raw data from each figure are provided in the source data. The data containing the results of all calculations generated in this study have been deposited in the Zenodo database under the accession code <https://doi.org/10.5281/zenodo.13300712>. TRNSYS code and models used to generate the results reported in this study are available upon request from the lead contact.

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AUTHOR CONTRIBUTIONS

Conceptualization, Y.Z. and L.Z.; methodology, Y.Z. and L.Z.; software, L.Z.; formal analysis, L.Z.; writing – original draft, L.Z.; writing – revision, Y.Z.; visualization, L.Z.; supervision, Y.Z.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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